FlashSVG: Fast and Faithful SVG Generation via Block Diffusion

Anonymous CVPR submission

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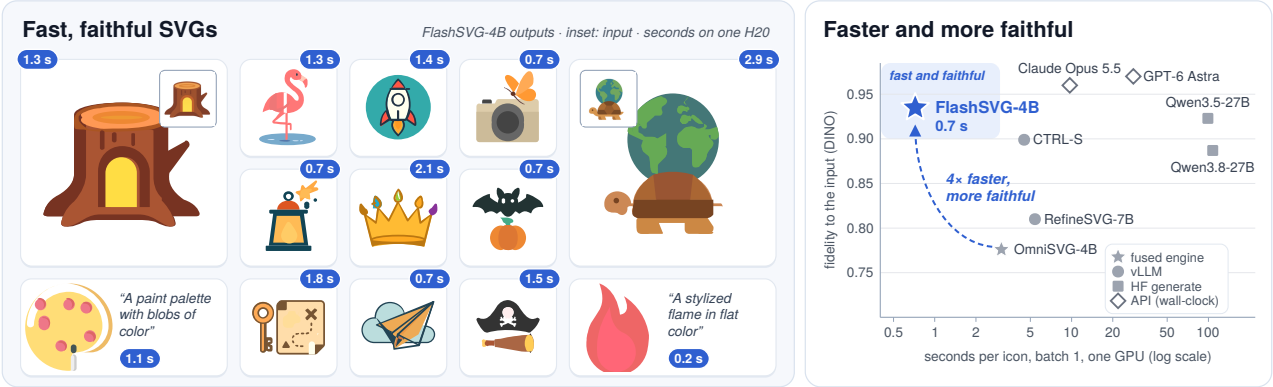


Figure 1. FlashSVG is fast and faithful. Left: FlashSVG-4B outputs picked by us for visual appeal from batch-1 runs of our fused decoder on one H20, each with its measured seconds (image-to-SVG with the input in the corner, text-to-SVG with the prompt). Means over the first 40 MMSVG-Bench cases per kind: 0.72 / 1.10 s per image-to-SVG icon / illustration and 1.0 / 3.3 s per text-to-SVG icon / illustration, against 3.07 / 5.23 s and 2.3 / 6.0 s for the released OmniSVG-4B on the same engine (unselected outputs: Figure 4 and Supplementary Figure S22). Right: icon DINO (Table 1) against seconds per icon on the fastest engine we ran (API models: wall-clock); the arrow compares FlashSVG-4B with the released model on the same engine. vtracer is faster and more faithful, also at FlashSVG's file size when tuned (Supplementary Figure S19).

Abstract

Language-model SVG generators are slow, and many of their drawings never finish: they decode one coordinate token per forward pass and often never emit an end-of-sequence token, running to the length cap while the truncated programs still render and pass validity-filtered evaluation. We show that termination, not syntax, is the main failure of the released OmniSVG models. Termination-aware post-training teaches a model to end its drawings instead of running to the cap, learning from its own rendered candidates, with non-terminating ones as preference negatives and no external judge; it also improves StarVector-8B and Qwen3.8-27B. FlashSVG, to our knowledge the first SVG generator trained with a block-diffusion objective over SVG tokens, drafts several tokens per forward pass and verifies them with the same weights, so committed tokens follow the model's own autoregressive distribution up to floating-point rounding; faithfulness comes from training, speed from drafting. On MMSVG-Bench image-to-SVG, FlashSVG reconstructs icons and illustrations more faithfully than the released OmniSVG-4B on every fidelity metric we report, and 99.7% / 98.2% of its candidates terminate. On one GPU at batch 1 and on the same engine, icon latency falls from 3.1 s for the released model to 1.4 s with post-training and to 0.7 s with verified decoding. The pixel tracer vtracer remains more faithful and faster on these clean renders, and the strongest API models more faithful; text-to-SVG remains weak.

1. Introduction

Scalable vector graphics describe a drawing as a program of paths, coordinates, and colors, so an icon stays sharp at any size and an illustration can be recolored without redrawing it. Vision-language models generate such programs from images and captions, as specialized path tokens (OmniSVG [102]) or as SVG code (StarVector [75]), building on hierarchical, autoregressive and stroke-level vector representations [11, 83, 95]. To serve design workflows, a generator must match the request, finish the program, and answer fast enough for interactive use. The last two are coupled: au-

toregressive decoding pays one sequential forward pass per coordinate token, and a program that never terminates pays it up to the length cap. The problem is to generate complete, faithful SVG in fewer sequential steps. The pixel tracer vtracer is more faithful and faster on clean renders (Figure 1); we address language-model generators, which also take text.

Two gaps separate generators from this goal. The first is termination: a path-token sequence truncated at the cap still renders, so the first-valid-of-eight protocol scores it like a finished drawing, and the failure shows only when termination is measured on its own (Section 3). Render-based post-training [54, 74, 76, 86] optimizes what a drawing looks like and handles runaway generation, if at all, with length budgets or reward gates [76, 101], without measuring termination.

The second gap is acceleration. Block diffusion [5] can draft tokens for causal verification (TiDAR, S2D2, DFlash [16, 32, 53]), but for SVG the unit of commitment matters: a coordinate plausible on its own can disagree with the causal continuation of the accepted path, and whether unverified parallel commitment, which succeeds for structured 3D generation (Block3D [22]), transfers to vector geometry is open.

FlashSVG addresses both gaps with one set of weights. Continued training of the released OmniSVG-4B adds a block-diffusion mode with 16-token blocks, as in Fast-dVLM [93], and a shifted readout keeps the autoregressive mode in the same output head. Termination-aware post-training samples and renders the model's own candidates; a candidate becomes the target only when it terminates and reconstructs the reference nearly as well as the ground truth does, and an odds-ratio preference loss [36] contrasts it with non-terminating candidates of the same prompt.

At inference, the diffusion mode drafts a block and the causal mode verifies it with speculative acceptance and residual resampling [13, 46], so committed tokens follow the adapted checkpoint's autoregressive distribution up to floating-point rounding. Post-training changes what the model generates and verified decoding only how fast, and both serve speed (Figure 1): on our fused decoder, post-training lowers batch-1 icon latency from 3.1 s to 1.4 s under autoregressive decoding by stopping candidates on time, and verification lowers it to 0.7 s at about 2.2 tokens per forward pass (Figure 6b).

FlashSVG builds on block-diffusion conversion, odds-ratio preference optimization, and self-speculative decoding; our contributions are:

- **Diagnosis.** Termination, not syntax, is the main failure of the released OmniSVG models, and first-valid scoring hides it; on MMSVG-Bench we report termination apart from validity, against open, API and non-learned baselines.
- **Termination-aware post-training.** Render-scored self-targets with a preference loss on non-terminating negatives improve reconstruction on MMSVG-Bench and two further sets and termination on MMSVG-Bench, and transfer to StarVector-8B and, with text-to-SVG replay, Qwen3.8-27B (VinciCoder-7B gains termination only, from its own targets); in a second round, low-score terminating negatives lower fidelity in each of three seeds, whereas runaway negatives keep it.
- **Verified block-diffusion decoding.** The diffusion mode drafts and the causal mode of the same weights verifies, keeping the autoregressive distribution (up to rounding) and halving icon latency of the same weights on the same engine, while the quality gain comes from training, as a causal-only control shows; a Block3D-style edit-aware variant of unverified parallel decoding does not transfer to SVG.

We will release the code (training, verified decoding with the fused engine, and evaluation), the FlashSVG-4B and FlashSVG-8B weights, and the per-case outputs and scores behind every table and figure.

2. Related Work

SVG generation with language models. Learned vector generators predict command sequences [11, 57], path or stroke tokens [95, 100, 102, 103], or SVG code [15, 75, 85, 87, 99, 111], evaluated on generation, editing and understanding benchmarks [47, 116, 117]; reinforcement learning optimizes rendered, rubric or geometric rewards [18, 54, 74, 76, 88], and render-in-the-loop methods refine their own drafts [52, 86]. Loops and unfinished outputs are met with a biased closing tag (StarVector), a length budget (RLRF), a reward gate (Back2Struct) or only reported (DuetSVG) [75, 76, 101, 111]; we measure termination across model families, show that first-valid scoring hides it, and train against it directly (IR4RL [41] and Back2Struct are concurrent).

Termination, length, and preference learning. Likelihood-trained decoders degenerate into repetition [35, 38, 91] and can fail to terminate [62, 90], also from images [9, 27]. Length confounds reinforcement and preference learning [56, 68]: DAPO masks or penalizes truncated samples [106], unlikelihood training pushes down a model's own repetitions [91], and reference-free losses can reward rejected responses that stop early [31]. Our targets follow filtered self-training [24, 30, 94, 109] and our loss the reference-free ORPO and SimPO [36, 61] within preference optimization [66, 67, 73], with non-terminating candidates as negatives; like pass@k [17], first-valid scoring hides per-sample failures.

Diffusion language models and verified decoding. Discrete and masked diffusion [6, 58, 77] scales to language and

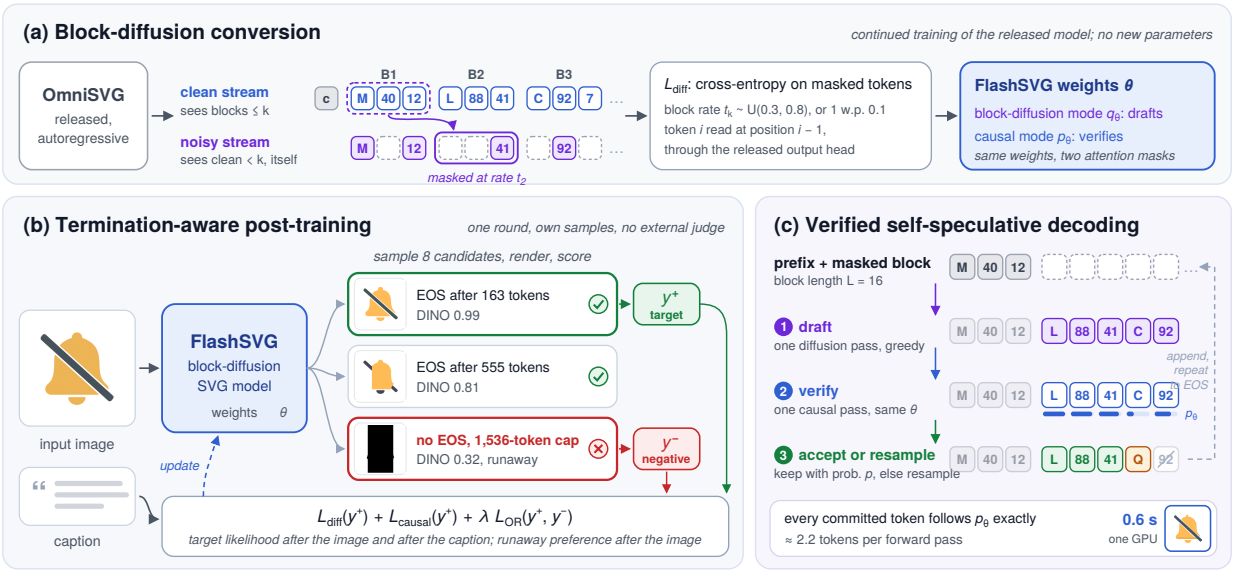


Figure 2. FlashSVG overview on a real MMSVG-Bench input. (a) Conversion: continued training of a released OmniSVG model runs a clean stream with block-causal attention and a noisy stream whose masked blocks see only themselves and earlier clean blocks; the diffusion loss is read out one position to the left through the released head, so the same weights θ have a block-diffusion mode q_θ and a causal mode p_θ. (b) Termination-aware post-training: the model renders and scores its own eight candidates; the best one that ends and scores at least 0.9 of the ground truth is the target y^+ (else the ground truth), a runaway one the negative y^- (Section 4.3); the loss adds an odds-ratio preference for y^+ over y^- (image prompt only) to the diffusion and causal likelihood of y^+ after the image and after the caption. (c) Verified decoding: one diffusion pass drafts a block, one causal pass of the same weights verifies it, and the first rejected token is resampled, so every committed token follows p_θ; the output shown took 0.6 s on one GPU.

vision-language models [63, 104, 105]; block diffusion interpolates with autoregression [5], and autoregressive models can be converted to it [20, 28, 93, 110]. Parallel decoding [12, 92] accelerates them but fails on strongly dependent outputs [42], whereas speculative decoding keeps the target distribution exact with separate drafters, draft heads, or diffusion drafters [10, 13, 16, 21, 46, 51, 82]; flat image-token distributions lower acceptance [40]. In SVG generation, diffusion has mostly entered through image priors or continuous vector latents [39, 84, 97, 98], and discrete-diffusion code models [29] can write SVG code without being trained for it. Closest to us, recent models draft with a diffusion mode and verify with a causal mode of the same weights [26, 32, 53, 78], also for coordinate outputs [107]. We credit the decoding rule to S2D2 and TiDAR; our evidence is specific to SVG: exact verification on vector paths, acceptance limited by coordinates, and the failure of unverified parallel decoding, including Block3D-style editing [22].

Vectorization. Per-image and learned vectorizers [34, 37, 48, 55, 59, 60, 69, 80, 113], text-guided optimization against image priors [25, 39, 98], and vector-space diffusion [84] produce SVG without a language model; two-stage pipelines [60, 96] vectorize a generated picture, like our cascade.

3. Termination Is the Bottleneck of the Released OmniSVG Models

Measuring termination apart from validity exposes the released model's dominant failure. A candidate is valid when its render is non-empty, and a sequence cut at the 1,536-token cap usually still renders (four times in five for the released OmniSVG-4B), so it counts as valid although it never finished (Figure 3).

Of the released OmniSVG-4B's image-to-SVG candidates, 37% of icons and 60% of illustrations never emit EOS (Table 1; 21% / 61% on text-to-SVG), whereas only 0.3% / 0.6% end without a renderable drawing, so termination, not syntax, is the failure; multi-candidate selection masks it: scored on its first sample, it reaches DINO 0.741 / 0.611, below its first-valid-of-eight scores, because the protocol replaces each failed sample with a later valid one. The second consequence is cost: a runaway candidate consumes the full token budget (48% of the released model's image-to-SVG candidates run to the cap, under 1% of FlashSVG-4B's; Figure 6a), so stopping on time is a speed-up, and an accelerated decoder must keep the model's stopping behavior, not merely produce renderable output.

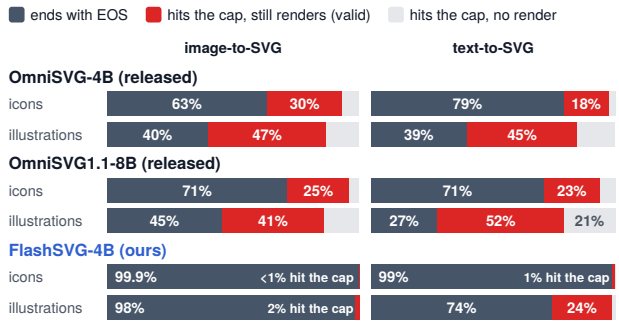


Figure 3. Termination is not validity (MMSVG-Bench, eight candidates per case; image-to-SVG left, text-to-SVG right). A candidate that hits the 1,536-token cap without EOS usually still renders and counts as valid under the official protocol (red). FlashSVG-4B candidates almost always end, except a quarter of text-to-SVG illustrations. A truncated SVG-code program does not render, so for Qwen3.5-27B validity tracks termination (87% / 72% of candidates end).

4. Method

4.1. Problem formulation

Let c be the condition, a raster image I or a caption, and $\mathcal{V}$ the OmniSVG path-token vocabulary [102] (about 45k tokens for commands, coordinates on a 200×200 grid, colors, and EOS). A drawing $x = (x_1, \ldots, x_T)$, $T \le T_{\max} = 1536$, comes from a causal generator

$$p_\theta(x \mid c) = \prod_{i=1}^{T} p_\theta(x_i \mid x_{<i}, c). \tag{1}$$

It is *terminating* when $x_T = \texttt{EOS}$ and *runaway* when it reaches $T_{\max}$ without EOS; a rasterizer $R(\cdot)$ renders any prefix, runaway or not. The goal is to raise the agreement of $R(x)$ with c together with the probability that x terminates, and to sample the resulting causal distribution exactly in fewer sequential forward passes, over blocks B_k of $L = 16$ tokens. FlashSVG does both with one set of weights θ (Figure 2): conversion adds a block-diffusion mode (a), termination-aware post-training reshapes the causal distribution (b), and verified decoding samples from it (c).

4.2. Shared-weight block-diffusion conversion

Block diffusion factorizes the sequence autoregressively over blocks, $p_\theta(x \mid c) = \prod_k p_\theta(B_k \mid B_{<k}, c)$, and applies masked diffusion within each block [5]. Each block draws its own masking rate $t_k \sim \mathcal{U}(0.3, 0.8)$, replaced by $t_k = 1$ with probability 0.1, and each of its tokens is masked with probability t_k, giving $\tilde{B}_k$ with masked positions M_k. With $\ell_{k,i} = -\log p_\theta(x_i \mid \tilde{B}_k, B_{<k}, c)$ and $M = \bigcup_k M_k$, the diffusion loss is the mean cross-entropy over the masked tokens of a sequence, averaged over sequences,

$$\mathcal{L}_{\text{diff}} = \mathbb{E}\Big[\frac{1}{|M|} \sum_k \sum_{i \in M_k} \ell_{k,i}\Big], \tag{2}$$

a flat weighting in place of the $1/t$ weighting of the masked-diffusion bound [77]. As in direct conversion of vision-language models [93], one forward pass processes a clean stream, the uncorrupted sequence under block-causal attention, and a noisy stream whose masked blocks attend bidirectionally to themselves and to the clean stream of preceding blocks only. The readout is shifted, as in Dream [104]: position i is predicted from the hidden state at $i-1$, so diffusion and next-token prediction share the released output head, and the same weights with causal attention define the autoregressive model $p_\theta(x_i \mid x_{<i}, c)$, from which verified decoding commits tokens while the diffusion mode only proposes them (q_θ). Conversion minimizes $\mathcal{L}_{\text{diff}}$ for 3,000 steps from OmniSVG-4B (a 3B vision-language backbone), which alone raises the causal model's icon DINO from 0.776 to 0.816 (Table 1). The causal loss $\mathcal{L}_{\text{causal}}$, the mean of $-\log p_\theta(x_i \mid x_{<i}, c)$ over the target tokens of a batch, enters in post-training; adding it to the conversion leaves image-to-SVG unchanged but lets nearly all text-to-SVG illustrations end (Supplementary Section S2).

EOS distillation at 8B. With $\mathcal{L}_{\text{diff}}$ alone, the 8B loses termination (Section 6.1). At every target position after a color token, where a path group can end, $\mathcal{L}_{\text{EOS}}$ is the Bernoulli KL divergence from the frozen released model's causal probability of EOS to the converted model's; the 8B conversion minimizes $\mathcal{L}_{\text{diff}} + 20\,\mathcal{L}_{\text{EOS}}$ for 1,000 steps; the 4B does not need it.

4.3. Termination-aware post-training

Post-training modifies p_θ itself, so we evaluate it with causal decoding before acceleration. For each of 14,000 training sources (an image I and its caption c_{txt}), the model samples eight candidates from the image prompt, renders them, and scores each against I:

$$\begin{aligned} s(y) = {} & 0.5\,\mathrm{IoU}_{\text{fg}}\big(R(y), I\big) \\ & + 0.5\big(1 - \mathrm{RMSE}_{\text{RGB}}(R(y), I)\big), \end{aligned} \tag{3}$$

with the foreground-mask overlap IoU_{fg} and the color error $\mathrm{RMSE}_{\text{RGB}}$ on a $[0, 1]$ scale. The positive target y^+ is the highest-scoring candidate that terminates and scores $s(y) \ge 0.9\, s(y_{\text{gt}})$, otherwise the ground truth; self-targets replace the ground truth for 90% of icon and 46% of illustration prompts, so the model is never trained to end a poor reconstruction. The negative y^- is the lowest-scoring runaway candidate whenever one exists (83% of the 10,241 pairs), otherwise the lowest-scoring candidate if it scores at least 0.1 below the target; the other 3,759 sources train only the likelihood terms. Following ORPO [36], with the length-normalized

likelihood $P_\theta(y \mid c) = \exp\left(\frac{1}{|y|}\sum_i \log p_\theta(y_i \mid y_{<i}, c)\right)$ and $\mathrm{odds}_\theta(y) = P_\theta/(1-P_\theta)$,

$$\mathcal{L}_{\mathrm{OR}} = -\log\sigma\Big(\log\frac{\mathrm{odds}_\theta(y^+)}{\mathrm{odds}_\theta(y^-)}\Big), \tag{4}$$

and both generation modes keep training on the target after the image and after the caption:

$$\mathcal{L}_{\mathrm{post}} = \sum_{c\in\{I,\, c_{\mathrm{txt}}\}} \Big[\mathcal{L}_{\mathrm{diff}}(y^+ \mid c) + \mathcal{L}_{\mathrm{causal}}(y^+ \mid c)\Big] + \lambda\,\mathcal{L}_{\mathrm{OR}}(y^+, y^- \mid I). \tag{5}$$

The preference term uses only the image prompt, from which the candidates were sampled and against which they were scored, so text-to-SVG receives image-grounded targets but no caption-level signal. A round has 400 updates over 48 sources each. λ was swept over eight values from 0 to 100 on the image-to-SVG cases of a validation split (MMSVG-val-600, held out from training, disjoint from MMSVG-Bench), where $\lambda = 30$ gives the highest illustration DINO (0.873, three seeds) and is within 0.001 of the best icon value; the final recipe uses $\lambda = 30$ for one round, and ablation arms A and B use $\lambda = 0$ (targets only) and $\lambda = 0.3$. The choice of negative follows a hypothesis about `EOS` in the odds ratio: a terminating but low-scoring negative ends with the same token as the positive, so pushing its likelihood down may also push down the probability of ending at all, whereas a runaway negative differs from a good target exactly in the property to suppress.

4.4. Verified self-speculative block-diffusion decoding

Decoding commits tokens only through p_θ, following the self-speculative pattern of S2D2 and TiDAR [32, 53] and speculative-sampling acceptance [13, 46]. Given the committed prefix $x_{<j}$, one diffusion pass fills a fully masked block with greedy drafts $\hat{x}_j, \ldots, \hat{x}_{j+L-1}$, point-mass proposals $q_i = \delta_{\hat{x}_i}$, and one pass of the same weights, strictly causal within the block as well, evaluates $p_i(\cdot) = p_\theta(\cdot \mid x_{<j}, \hat{x}_{j:i-1}, c)$ at all draft positions in parallel. Scanning from left to right, draft $\hat{x}_i$ is accepted with probability $\min(1, p_i(\hat{x}_i)/q_i(\hat{x}_i)) = p_i(\hat{x}_i)$. At the first rejection, a replacement is sampled from the normalized residual $(p_i - q_i)_+$, which is p_i with $\hat{x}_i$ removed, the rest of the block is discarded, and the key-value cache is truncated; if all L drafts are accepted, a bonus token is sampled from the final conditional (Supplementary Algorithm 1). Here p_i is what autoregressive sampling of the checkpoint draws from (SVG token set; official temperature, top-k, top-p and repetition penalty over the committed and earlier draft tokens, Supplementary Table S1), and a block crossing the token cap is cut there. In exact arithmetic, the speculative-sampling argument [46] then guarantees that each committed token follows $p_\theta(\cdot \mid x_{<i}, c)$, the converted checkpoint's causal distribution (tested numerically in Section 6.3); it says nothing about the released model's outputs or visual correctness.

Cost is governed by forward passes: τ counts committed tokens per forward pass, draft and verification passes included ($\tau = 1$ for autoregressive decoding), and since a speculative pass processes L positions, speedup is τ divided by the ratio of per-pass costs. Unverified decoders skip verification, no longer sample from p_θ, and form a separate class.

5. Experiments

5.1. Benchmarks and metrics

MMSVG-Bench has 150 cases for each combination of icon or illustration and image-to-SVG or text-to-SVG, reported separately. Two image-to-SVG sets test robustness: 300 out-of-domain ColorSVG-100K icons [19] and a stress set of 92 complex scenes (median 143 elements, mostly beyond the token cap), each split at its median element count into a simple and a complex half. No MMSVG-Bench input has a near-duplicate among our 1.16M training renders (DINOv2 search, Supplementary Section S1). Image-to-SVG is scored by the cosine similarity of DINOv2 CLS embeddings [65] between the 224-px render on white and the input, with SSIM [89] and LPIPS [112] at 256 px as pixel-level and perceptual checks; text-to-SVG by CLIP ViT-B/32 similarity [33, 72] and a GPT-6 Astra judge that rates 512-px renders from 0 to 10 for prompt fidelity and visual quality [14, 115] (invalid outputs score 0; Supplementary Section S1). Following the official OmniSVG protocol [102], up to eight candidates are sampled per case and the first with a non-empty render is scored, zero if none; we also report one-sample scores, candidate and output EOS rates, path tokens, τ, and batch-1 seconds per case.

5.2. Baselines and implementation

Baselines are the released OmniSVG-4B and OmniSVG1.1-8B [102], six open SVG generators [75, 87, 88, 100, 114], four Qwen vision-language models [70, 71], Qwen3.8-27B fine-tuned by us on GPT-6 Astra SVGs, eight API models, and four non-learned vectorizers, each with its released settings (Supplementary Section S1). We do not compare separately trained draft heads [10, 49–51], which add parameters and training. FlashSVG-4B, our main model and FlashSVG without a size, is trained from OmniSVG-4B, and FlashSVG-8B from OmniSVG1.1-8B with EOS distillation in a conversion stopped at step 1,000; post-training is identical. Latency is measured on one H20 at batch 1. Main arms use three training seeds (mean and min–max range), each evaluated once. Differences are paired per case, with per-case scores averaged over seeds, 95% bootstrap inter-

Table 1. MMSVG-Bench [102] under the official protocol (up to eight candidates; first valid scored; invalid = 0; Supplementary Table S7 lists ten further systems). Image-to-SVG: DINO, SSIM and LPIPS [112] against the input (invalid: SSIM 0, LPIPS 1), EOS = share of candidates that terminate, s = batch-1 seconds per case on one H20 (HF generate; other engines in Supplementary Table S9). FlashSVG: candidates sampled autoregressively, which verified decoding matches up to rounding (Section 6.3); s with verified decoding in our HF-based harness. Text-to-SVG: CLIP and judge fidelity (GPT-6 Astra, 0–10). API models: one sample, lowest reasoning setting, 24k-token cap, on the input's 448-pixel canvas, as general vision-language models. Baselines are single evaluations; the two FlashSVG rows and OmniSVG1.1-8B + post-training are means of three training seeds. Bold: best open-weight learned value (not for s, whose engines differ). [a]Remote wall-clock. [s]Timed on one seed. [d]Fine-tuned by us on GPT-6 Astra SVGs. [v]vLLM. [c]CPU. [e]Conversion with EOS distillation (Section 4.2).

	Icons							Illustrations						
System	**DINO↑**	**SSIM↑**	**LPIPS↓**	**CLIP↑**	**Judge↑**	**EOS↑**	**s↓**	**DINO↑**	**SSIM↑**	**LPIPS↓**	**CLIP↑**	**Judge↑**	**EOS↑**	**s↓**
Closed models (API)														
GPT-6 Astra [64]	0.970	0.972	0.021	0.308	9.79	–	28.3[a]	0.981	0.984	0.009	0.313	9.83	–	35.1[a]
Claude Opus 5.5 [4]	0.960	0.890	0.057	0.304	9.71	–	9.7[a]	0.939	0.900	0.061	0.316	9.44	–	12.7[a]
Open-weight general model (API)														
Kimi K3 [43]	0.922	0.880	0.092	0.304	8.83	–	24.6[a]	0.895	0.872	0.109	**0.306**	8.57	–	27.6[a]
Non-learned image-to-SVG														
vtracer [69]	0.982	0.987	0.009	–	–	–	0.02[c]	0.978	0.987	0.009	–	–	–	0.02[c]
LIVE [59]	0.945	0.994	0.006	–	–	–	792	0.955	0.994	0.006	–	–	–	779
AdaVec [113]	0.935	0.979	0.024	–	–	–	83	0.947	0.986	0.018	–	–	–	80
Open general vision-language models														
Qwen3.5-27B [70]	0.923	0.873	0.093	0.289	6.36	0.87	99.1	0.833	0.812	0.172	0.294	7.33	0.72	110.6
Qwen3.8-27B [71]	0.887	0.804	0.171	**0.307**	8.87	0.87	107.6	0.809	0.763	0.226	**0.306**	8.77	0.69	135.1
Open SVG models														
Qwen3.8-27B + GPT-6 Astra SFT[d]	0.889	0.767	0.226	0.303	**8.94**	0.97	67.5	0.872	0.777	0.245	0.298	**8.99**	0.96	77.2
InternSVG-8B [87]	0.863	0.819	0.161	0.282	5.67	0.81	–	0.840	0.842	0.155	0.157	1.89	0.59	–
CTRL-S (8B) [88]	0.899	0.822	0.142	0.295	6.92	**1.00**	15.1	0.854	0.836	0.159	0.276	5.89	**1.00**	20.1
StarVector-1B [75]	0.896	0.812	0.143	–	–	0.77	15.7	0.698	0.669	0.315	–	–	0.46	27.2
VinciCoder-7B [114]	0.781	0.654	0.505	–	–	0.92	12.6[v]	0.746	0.698	0.499	–	–	0.78	30.2[v]
OmniSVG-4B [102]	0.776	0.837	0.197	0.271	4.46	0.63	17.3	0.707	0.834	0.245	0.215	1.35	0.40	26.9
OmniSVG1.1-8B [102]	0.800	0.848	0.188	0.270	4.87	0.71	12.8	0.690	0.831	0.262	0.218	1.43	0.45	24.6
Ours														
OmniSVG1.1-8B + post-training	0.869	0.885	0.122	0.270	5.14	0.97	8.2[s]	0.825	0.882	0.148	0.223	2.51	0.91	11.3[s]
FlashSVG-8B[e] (λ=30)	0.914	0.909	0.088	0.274	5.49	0.981	4.5	0.884	0.910	0.101	0.229	2.75	0.925	6.8
OmniSVG-4B + block-diff. conversion	0.816	0.862	0.159	0.274	–	0.68	–	0.728	0.862	0.211	0.214	–	0.39	–
FlashSVG-4B (λ=30)	**0.935**	**0.928**	**0.064**	0.276	5.42	0.997	4.3	**0.904**	**0.918**	**0.083**	0.223	2.54	0.982	6.1

vals over cases (10,000 resamples) and sign-flip permutation p-values [7, 44]. One evaluation run of a fixed checkpoint varies by about ±0.006 / ±0.009 DINO (icons / illustrations), and we do not interpret smaller single-run differences.

6. Results

6.1. Main comparison on MMSVG-Bench

Among open learned models evaluated under the official protocol, FlashSVG-4B attains the highest image-to-SVG DINO, SSIM and LPIPS and, of those timed, the lowest latency (Tables 1 and S9). Post-training alone also improves the released 8B in every column of Table 1 (three seeds; DINO 0.800 / 0.690 to 0.869 / 0.825, candidate termination 0.71 / 0.45 to 0.97 / 0.91). Conversion contributes beyond post-training at both sizes: without it, the same post-training gives 0.877 / 0.756 at 4B, so conversion adds +0.059 / +0.148 DINO at 4B and, at 8B, +0.014 / +0.015 with the 4B recipe (intervals include zero) and +0.046 / +0.059 with EOS distillation (three seeds per cell; Supplementary Table S2b). At 4B this gain is continued SVG training, not the diffusion objective: the same 3,000 updates with the causal loss alone, post-trained as FlashSVG-4B, give 0.935 / 0.896 (difference 0.000 [−0.014, 0.018] / −0.008 [−0.018, 0.003]; Supplementary Section S2), so the diffusion objective serves drafting.

At 8B, conversion with the recipe of FlashSVG-4B lowers icon DINO and termination, and EOS distillation keeps termination where freezing the untied output layer does not (Supplementary Table S2b, Section S2). With it, FlashSVG-8B reaches 0.914 / 0.884 (+0.031 [0.014, 0.049] / +0.044 [0.028, 0.060] over the 4B recipe), and verified decoding is 1.73 / 1.63× faster than autoregressive decoding of the same weights (4.5 / 6.8 against 7.8 / 11.0 s, three seeds); the 8B still trails FlashSVG-4B (paired −0.021 [−0.032, −0.011] / −0.020 [−0.033, −0.007]), which remains our main model.

Among open baselines, Qwen3.8-27B distilled from GPT-6 Astra is among the strongest but much slower, and CTRL-S [88] terminates every candidate but is slower on either engine (4.5 / 6.1 s on vLLM; all baselines in Supplementary Table S7). API models lead text-to-SVG, and on image-to-SVG GPT-6 Astra leads every learned model in every column. vtracer is more faithful on these clean renders,

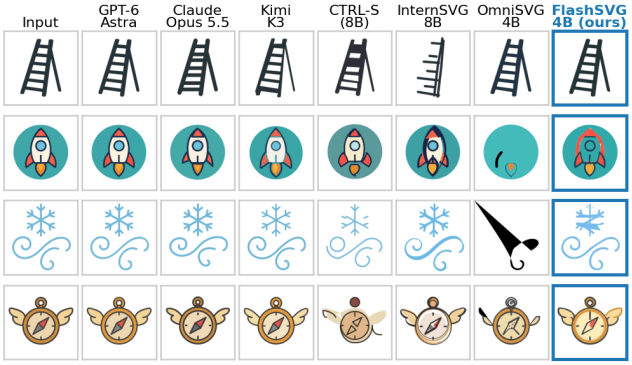


Figure 4. MMSVG-Bench image-to-SVG, the first two icons and illustrations by id (not selected): the scored output of each system (first valid of eight; one sample for API models, on the pixel canvas).

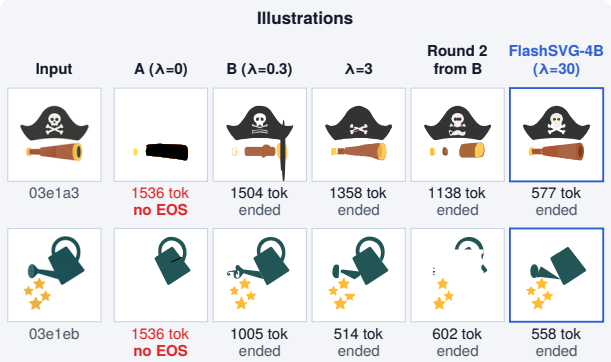


Figure 5. Illustrations whose scored output from round-1 arm A (λ=0) never ends while that of FlashSVG-4B (λ=30) does, the first two by id (icons and more cases: Supplementary Figure S7). Columns: the input, then arms A, B (λ=0.3), λ=3, round 2 from B and FlashSVG-4B, each with its generated SVG tokens and EOS status (red: cut at the 1,536-token cap).

even tuned to FlashSVG's mean file size (DINO 0.951 / 0.960 against 0.935 / 0.904; Supplementary Figure S19), so FlashSVG's case rests on learned, conditional generation rather than on compressing renders. FlashSVG's one-sample DINO equals its first-valid score (training seed 1; one-sample validity 0.99 / 1.00), so its quality does not depend on resampling. Figure 4 shows unselected cases (selected and random ones in Supplementary Figures S5 and S21).

Text-to-SVG is FlashSVG's weakest task, as expected from a preference signal that sees only the image: its fidelity trails every API model, CTRL-S, the 27B Qwen models and, on one kind, four other open models (Tables S7 and S8), although post-training raises judge fidelity over the released OmniSVG-4B from 4.46 / 1.35 to 5.42 / 2.54 and candidate termination from 0.79 / 0.39 to 0.98 / 0.72. Vectorizing text-to-image pictures and picking with a local judge reaches judge fidelity 8.60 / 7.43 at an estimated 200–260 GPU-seconds per prompt (Supplementary Section S5).

6.2. Termination-aware post-training

Table 2 isolates the components.

Self-generated targets alone (λ=0) improve on conversion, and the (mostly runaway) negatives add most of the gain: relative to λ=0, λ=30 gains 0.082 [0.060, 0.105] / 0.158 [0.133, 0.184] DINO ($p < 0.001$), termination rises from 0.78 / 0.53 to 0.99 / 0.99, and icon outputs shorten from 651 to 430 tokens while leaving less of the input undrawn: 3% / 4% of its ink for FlashSVG-4B, 21% / 24% for the released model (Figure 5; Supplementary Section S4). Gains saturate between λ=10 and 30; the main-table row, sampled autoregressively, is within the seed range. The negative type matters: in a second round from FlashSVG-4B (same pool, targets and updates), pairing targets with low-score candidates that end (96% of pairs, as the model rarely runs away) lowers termination and DINO in each of three seeds, whereas runaway negatives alone leave both intact, consistent with the shared-EOS hypothesis but not isolating it, since the arms also differ in pair count. Once termination is solved, more training does not help: a ground-truth-only arm (0.847 / 0.745) performs like λ=0, extra rounds change MMSVG-Bench insignificantly, and render-reward GRPO [81] collapses without ratio clipping and leaves FlashSVG-4B unchanged with PPO-style clipping [79] (p = 0.56 / 0.84). The recipe transfers to StarVector-8B (0.729 / 0.520 to 0.836 / 0.802) and, with text-to-SVG replay, to Qwen3.8-27B (0.887 / 0.809 to 0.934 / 0.899 on the pixel canvas), with termination rising in both; VinciCoder-7B needs targets in its own style, and n-gram bans or a presence penalty do not substitute for post-training (Supplementary Tables S10 and S11).

6.3. Verified decoding and the parallel-decoding negative result

On the final weights, verification keeps termination and cuts latency from 8.5 / 11.1 s to 4.3 / 6.0 s per case and the fused engine to 0.72 / 1.10 s (first 40 cases per kind), against 3.07 / 5.23 s for the released model on the same engine and 2.5 / 3.6 s for the same weights on vLLM (Figure 6b; Supplementary Tables S3 and S9). Token by token the decoders sample the same distribution (total variation at most $6 \cdot 10^{-5}$ in float32, 99th percentile). Over three to five sampling runs per decoder and the three post-trained models, verified decoding scores +0.005 DINO above autoregressive sampling ([0.000, 0.009] / [0.001, 0.009]), but +0.003 / +0.002 without the runs that prompted the repeats and +0.001 / $-$0.002 in float32, intervals including zero; the fused engine's outputs match in length, termination and DINO (Supplementary Sections S3 and S6). Drafts break at coordinates (0.886 / 0.899 of first mismatches), and iterative threshold drafting, drafter alignment and prompt-lookup drafts do not help (Supplementary

Table 2. Post-training ablations under verified decoding (image-to-SVG DINO, first valid of eight; conversion only: autoregressive, as in Table 1): components and preference weight λ of one round from the converted OmniSVG-4B, and the negative type of a second round from FlashSVG-4B. Three-seed means with min–max ranges; †single runs. ColorSVG-100K is out of domain, "Stress" the 92-scene set; EOS_{out}: share of scored outputs that terminate on MMSVG-Bench (Table 1 counts candidates), τ: tokens per forward pass (icons / illustrations). Bold: best among one-round arms.

	MMSVG-Bench		ColorSVG-100K		Stress			
Arm	**Icons**	**Illus.**	**Simple**	**Complex**	**Simple**	**Complex**	**EOS_{out}**	τ
Conversion only†	0.816	0.728	0.754	0.627	0.227	0.165	–	–
+ self-targets, A (λ=0)	0.855 [0.845, 0.873]	0.752 [0.732, 0.776]	0.772†	0.655†	0.233†	0.164†	0.78 / 0.53	**3.00 / 3.13**
+ negatives (83% runaway), B (λ=0.3)	0.911 [0.908, 0.914]	0.849 [0.843, 0.860]	0.832†	0.760†	0.385†	0.272†	0.95 / 0.84	2.40 / 2.27
λ=3	0.930 [0.928, 0.931]	0.881 [0.867, 0.888]	0.862 [0.859, 0.869]	0.810 [0.802, 0.816]	0.436 [0.433, 0.439]	0.340 [0.326, 0.356]	0.98 / 0.92	2.33 / 2.14
λ=10	0.932 [0.928, 0.936]	0.897 [0.894, 0.900]	0.879†	0.817†	0.475†	0.382†	0.98 / 0.97	2.28 / 2.07
λ=100†	0.932	0.899	0.865	0.805	**0.529**	**0.440**	**0.99 / 0.99**	2.14 / 1.89
FlashSVG-4B, λ=30	**0.937** [0.933, 0.939]	**0.910** [0.903, 0.918]	**0.880** [0.876, 0.887]	**0.827** [0.824, 0.829]	0.506 [0.495, 0.512]	0.409 [0.390, 0.423]	**0.99 / 0.99**	2.22 / 2.03
Second round from FlashSVG-4B: negative type								
Runaway negatives only	0.940 [0.933, 0.948]	0.902 [0.900, 0.903]	–	–	–	–	0.99 / 0.98	–
Negatives as in round one (96% low-score)	0.807 [0.774, 0.827]	0.716 [0.683, 0.743]	–	–	–	–	0.78 / 0.62	–

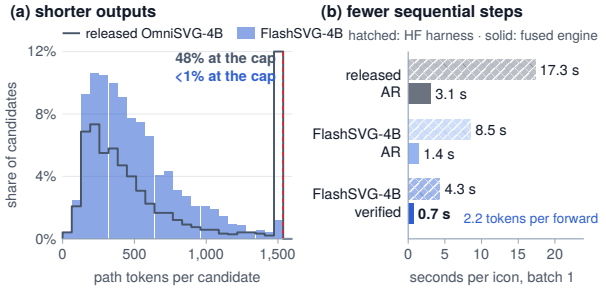


Figure 6. Where FlashSVG's speed comes from (MMSVG-Bench image-to-SVG). (a) Path tokens per candidate (eight per case, training seed 1): 48% of the released model's candidates run to the 1,536-token cap, under 1% of FlashSVG-4B's. (b) Batch-1 seconds per icon case on one H20, in the HF-based harness of Table 1 (hatched) and on our fused engine (solid); verified decoding commits 2.2 tokens per forward pass.

Section S3). Unverified commitment fails outright: parallel decoding yields renderable drawings with low scores, and our Block3D-style edit-aware training [22] (from B, up to 1,000 steps) makes it worse, lowers termination, and slows generation (Supplementary Table S3).

7. Conclusion and Limitations

FlashSVG is post-trained against runaway generation and decoded with same-weight verified speculation; among open learned models it reconstructs images best and, of those timed, at the lowest latency, while the strongest API models and vectorizers (even at equal file size) remain more faithful on these clean renders. Post-training and verified drafting are separable: the first makes drawings more faithful, terminating and shorter, the second only lowers latency.

Only λ was chosen on a validation split; MMSVG-Bench informed other choices (e.g., rounds and the 8B's EOS-distillation settings), so it is not fully held out (on cases no check used, the ranking holds), while the out-of-domain sets informed none (Supplementary Section S1). Holm's correction over the main text's 16 tests makes only the +0.005 of verified decoding non-significant (adjusted p = 0.34 / 0.22), and resampling seeds and cases moves no interval across zero (Supplementary Section S1); each seed was evaluated once, and the causal-only control was run at 4B only. DINO, SSIM and LPIPS are not human ratings, and the GPT-6 Astra judge is also the closed reference, though Claude Opus 5.5 as a second judge ranks the text-to-SVG rows alike (Kendall τ = 0.94 / 0.97; Table S5). Text-to-SVG remains weak, most stress scenes exceed the token cap, the 8B (a different release, OmniSVG1.1) needs EOS distillation and trails the 4B, and verified decoding runs only in our decoders, whose gain shrinks at large batch (Tables S4 and S9).

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Supplementary Material

S1. Implementation details

Table S1 lists the settings used for conversion, post-training, and evaluation.

Baselines, hardware, and timing. Failed judge calls are retried; the five that failed on every retry (three icons and two illustrations of one FlashSVG-8B training seed) are left out. The learned baselines are InternSVG-8B [87], CTRL-S [88], StarVector-1B and 8B [75], VinciCoder-7B [114] and HiVG-3B [100], each with its released prompt, sampling, token budget and decoder; the general vision-language models Qwen3.5-4B, 9B and 27B [70] and Qwen3.8-27B [71]; Qwen3.8-27B fine-tuned by us on GPT-6 Astra SVGs; and eight API-served models (five closed, three open-weight; one sample each). OmniSVG-family path-token models are capped at 1,536 tokens and the Qwen models receive 4,096 new tokens; Table S8 adds further learned baselines and a second benchmark. The four non-learned vectorizers process each image individually with a single setting: vtracer uses its defaults, DiffVG [48] optimizes 256 randomly initialized paths for 500 Adam steps on MSE, LIVE [59] runs its official 128-path schedule, and AdaVec [113] runs the one configuration of its official code (SAM and superpixel regions, 500 joint iterations). Training and evaluation ran on 16 hosts with eight H20 GPUs each. Latency covers the scored output; CTRL-S and VinciCoder-7B are timed on the first 40 cases per kind, the general vision-language models, StarVector and HiVG-3B on the first 10 (Section S6), and the OmniSVG family in our HF-based harness on all 150; an API request retried after an empty reply is timed on its retry; API timings are remote wall-clock times under concurrent requests and are not ranked; the released model's own batch-8 script takes 25.2 / 29.5 s per case. Our training code builds on the public DiffusionVL code [110].

Data and development decisions. Training data are the MMSVG-Icon and MMSVG-Illustration releases [102] (1,159,423 SVGs in the path-token format); a random 0.1% validation split (1,136 SVGs, no token, SVG or image hash shared with training, near-duplicates aside) yields MMSVG-val-600. Conversion trains on the training split, and post-training draws its 14,000 sources from it. λ (eight values, three seeds each) was chosen on MMSVG-val-600. MMSVG-Bench informed the target threshold (0.8 or 1.0 instead of 0.9 changes DINO by at most 0.012, one run each at λ=0.3), the number of rounds, the decoder, and at 8B the EOS-distillation weight (1 or 20) and stopping step (500 or 1,000), the last two on its first 40 cases per kind. Block length 16 and 3,000 conversion steps at 4B were fixed in advance (step 1,000 gives the same end-to-end result, 0.931 / 0.903). ColorSVG-100K and the stress set informed no choice. Differences in the text are computed from unrounded means, so they can differ by 0.001 from differences of printed values. On the 110 cases per kind that no check used (ids 41–150), the ranking of Table 1 is unchanged: FlashSVG-4B 0.930 / 0.907, FlashSVG-8B 0.906 / 0.884 (+0.045 [0.026, 0.063] / +0.050 [0.029, 0.072] over post-training without conversion), CTRL-S 0.902 / 0.862.

Train/test overlap. We embed all 1,158,287 training renders and every evaluation input with DINOv2-base (as the metric) and call an input a near-duplicate when its nearest training render has cosine ≥ 0.95 and SSIM ≥ 0.9. As calibration, a training render re-rendered at 224 px has cosine ≥ 0.989 and SSIM ≥ 0.990 to its original in 95% of 300 cases. No MMSVG-Bench input is a near-duplicate (largest cosine 0.981, for a different drawing of the same concept), and no training caption matches a text-to-SVG prompt (largest TF-IDF cosine 0.87). SArena-600 is not clean: 14 of its 150 icons and 31 of its 150 illustrations are near-duplicates, and 3 of its 300 prompts match a training caption (Table S8); ColorSVG-100K has one, the stress set none, and the in-distribution MMSVG-val-600 91; on its 209 other cases, λ=30 still gives the highest illustration DINO (0.866) and is within 0.004 of the best icon value (one run of λ=100).

Reproducibility. With the per-case outputs, judge scores and candidate records we will release a script that recomputes printed numbers from them, rounding unrounded means half up; it currently covers every numeric cell of Tables 1 to S3, of Tables S5 to S11 and of Table S2b, 34 sentences of the main text and this supplement, and seven statements on intervals, Holm's correction, the pre-registered rule and the cascade's selection rule, without a mismatch; it does not yet cover the remaining numbers, mostly in the supplementary text. Re-scoring with the metric of Table 1 is bit-identical (five evaluations, three of them of FlashSVG-4B, scored two days apart give the same means); with that of Table 2, the one run we re-scored ten days later (runaway negatives only, first seed) differs by at most $1.3 \cdot 10^{-4}$ per case and $2.4 \cdot 10^{-6}$ in its means, which leaves one cell of Table 2 (0.940, the same arm) within 10^{-6} of a rounding boundary. Regenerating an evaluation with the same seeds on another H20 reproduces its scored outputs byte for byte (Section S3). We did not repeat a training run with identical seeds; three training seeds per arm measure the spread of retraining, and one evaluation run of a fixed checkpoint varies by about $\pm$0.006 / $\pm$0.009 DINO (Section 5.2). Judge scores come from an API model and are released rather than re-queried.

Multiplicity. We apply Holm's correction to three nested families of two-sided tests, icons and illustrations separately (sign-flip p-values $(1 + b)/(N + 1)$ with N = 10,000 flips;

for the pooled verified-minus-autoregressive estimates, the normal p of their z). (i) The 16 tests behind the intervals and p-values of the main text: 8 have $p < 0.05$, and all of them except the +0.005 of verified decoding (adjusted p = 0.34 / 0.22) stay below 0.05 (largest adjusted p 0.027, FlashSVG-8B against FlashSVG-4B on illustrations). (ii) Adding the effects the text states without an interval (what conversion adds at 4B and 8B, post-training alone at 8B, FlashSVG-4B over OmniSVG-4B, and the causal-only updates) gives 28 tests; the same two drop, and conversion with the 4B recipe at 8B is not significant even uncorrected (p = 0.18 / 0.11). (iii) Over all 150 paired tests we ran on MMSVG-Bench, ColorSVG-100K and the stress set, 56 of the 84 with $p < 0.05$ remain, including 14 of the 16 that survive (ii); the other two are the 8B's icon gain over the 4B recipe (adjusted p = 0.066) and its illustration gap to FlashSVG-4B (0.22). A bootstrap that resamples the training seeds of each arm as well as the cases widens every interval of (ii) but moves none across zero.

Figure S1 summarizes the final model across all regimes.

S2. Conversion at 8B and the causal term

With the recipe of FlashSVG-4B, conversion degrades the causal decoding of OmniSVG1.1-8B instead of improving it (Table S2a). Within 500 steps the share of outputs that end falls from 0.78 / 0.57 to 0.50 / 0.28, and icon outputs lengthen from 716 to 1,012 tokens on average; neither recovers by step 2,000. On icons, most of the additional unfinished outputs are still drawing new paths at the 1,536-token cap: loops, in which at least half of the paths in the second half of the output repeat an earlier path exactly, stay at 4–5 of the 40 cases while unfinished outputs rise from 9 to 16–20. On illustrations, loops (8 to 12–16) and unfinished drawings both increase. Teacher-forced on 1,328 held-out references (1,084 icons, 244 illustrations: the 192 evaluation-cohort sources left out of the training data and the validation split), conversion lowers the per-token NLL at both sizes (8B: 0.214 / 0.702 to 0.145 / 0.517 at step 3,000; 4B: 0.331 / 0.797 to 0.253 / 0.639) and leaves EOS the top prediction at the true end of nearly all icons. What changes is the probability of EOS after the earlier path groups, where a free-running model that has finished its own drawing has to stop: it falls by 48% / 68% at 8B (from 10.8 / 21.6 to 5.6 / 6.9 $\times 10^{-3}$) but by 6% / 28% at 4B, and adding the causal term to the conversion does not stop the fall (7.7 / 9.9 $\times 10^{-3}$ at step 1,000, against 7.7 / 8.9 without it). Neither adding $\mathcal{L}_{\text{causal}}$ (weight 1) to the conversion, although its loss on the training data keeps falling, nor halving the learning rate changes this; both runs were stopped after the last check shown. At 4B the same causal term raises termination during conversion. Post-training restores termination at 8B but not the fidelity of FlashSVG-4B (Table S2b).

Table S1. Implementation and evaluation settings.

Setting	Value
Model	
Base model	OmniSVG-4B (3B vision-language model)
Block length	16
Conversion steps	3,000 (8B: 1,000)
Conversion optimizer	AdamW, peak LR 10^{-5}, 100 warmup updates, cosine over 6,000 updates, 192 sequences per update, bf16
Post-training	
Rounds (final model)	1
Updates per round / sources per update	400 / 48
Optimizer	AdamW, peak LR 5×10^{-6}, cosine to 0.2×, clipping 1.0, all parameters
Loss reductions	$\mathcal{L}_{\text{diff}}$, $\mathcal{L}_{\text{causal}}$: token means; $\mathcal{L}_{\text{OR}}$: mean over pairs; end-token padding to whole blocks always masked, never scored
Training prompts / candidates per prompt	14,000 / 8
Target threshold	0.9 of the ground-truth score
Preference pairs	10,241
λ sweep	0, 0.1, 0.3, 1, 3, 10, 30, 100
Training seeds per main arm	3
Evaluation	
Sampling (temperature / top-p / top-k / repetition penalty)	official: image-to-SVG 0.3 / 0.90 / 50 / 1.05; text-to-SVG icons 0.5 / 0.88 / 50 / 1.05, illustrations 0.6 / 0.90 / 60 / 1.03
Token cap	1,536 path tokens; 4,096 new tokens for general code models; released budgets for SVG code models; 24k for API models
Render size	DINO 224 px; SSIM, LPIPS 256 px; judge 512 px
Bootstrap resamples	10,000

EOS distillation and the output layer. Distilling the released model's probability of EOS after color tokens keeps termination through the 8B conversion (Table S2a): at step 1,000, 0.85 / 0.42 of the outputs end, against 0.60 / 0.22 without the term (on all 150 + 150 cases 0.73 / 0.47 against 0.54 / 0.28), while weight 1 is too weak (0.60 / 0.32). The untied output layer moves most during the 8B conversion (relative weight change $52.6 \cdot 10^{-3}$ against $22.8 \cdot 10^{-3}$ for the language-model layers), but it does not carry the loss: frozen, or trained at a tenth of the learning rate, it leaves termination as low as without the term (0.48 / 0.18 and 0.50 / 0.32 at step 1,000). Freezing the vision tower raises icon DINO, but termination far less (0.65 / 0.32). Post-trained, the EOS-distilled conversion at step 1,000 reaches 0.914 / 0.884 (three seeds), against 0.883 / 0.840 with the 4B recipe (Table S2b); a second conversion seed gives 0.916 / 0.878 (one post-training seed) and the same conversion with the vision tower frozen 0.914 / 0.883 (three seeds). At 4B the

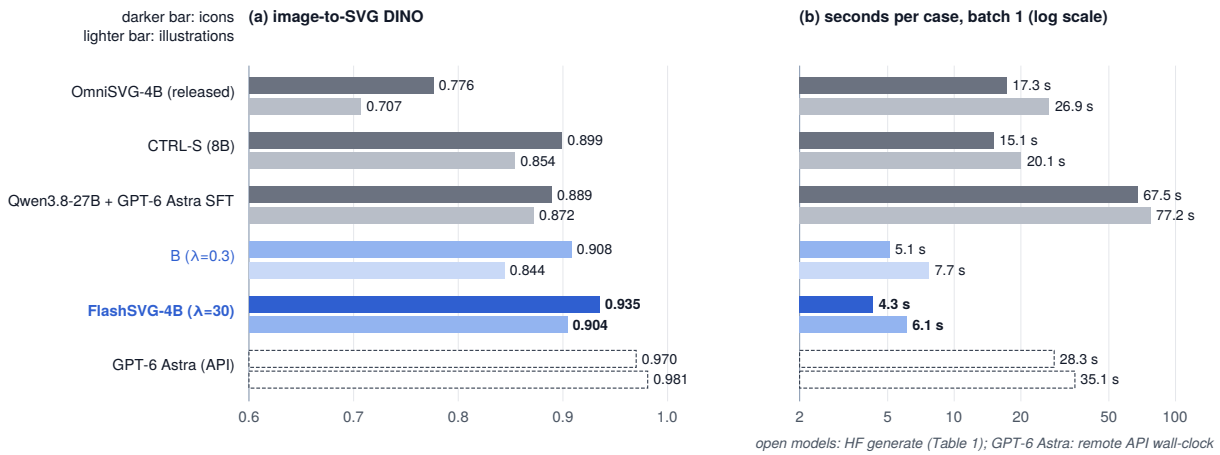


Figure S1. Summary of FlashSVG-4B against the released OmniSVG-4B, CTRL-S, Qwen3.8-27B with GPT-6 Astra SFT, and B (λ=0.3): (a) image-to-SVG DINO (first valid of eight; one sample for GPT-6 Astra) and (b) batch-1 seconds per case (HF generate for open models, remote wall-clock for GPT-6 Astra). All values are those of Table S7.

term is not needed: conversion keeps termination without it, and of two conversion seeds with it, one diverged after update 735 (gradient norm up to 338) and ends its outputs far too early, while the other stays close to the conversion without the term (0.820 / 0.641 against 0.813 / 0.707 at step 1,000).

Conversion with and without post-training. Post-training alone, applied to the released models with their own candidate pools and without $\mathcal{L}_{\text{diff}}$, gives 0.877 / 0.756 at 4B and 0.869 / 0.825 at 8B (Table S2b). Conversion before post-training adds +0.059 [0.035, 0.081] / +0.148 [0.124, 0.172] at 4B, +0.014 [−0.007, 0.035] / +0.015 [−0.004, 0.034] at 8B with the 4B recipe and +0.046 [0.030, 0.061] / +0.059 [0.041, 0.077] with EOS distillation (paired over cases); without conversion, the 8B is the stronger model on illustrations, with it the 4B is.

Causal-only control. At 4B a control separates the conversion objective from extra SVG training: the conversion of FlashSVG-4B repeated with $\mathcal{L}_{\text{causal}}$ in place of $\mathcal{L}_{\text{diff}}$ (same data, schedule, batch and 3,000 updates), its own round-1 pool, and post-training with three seeds, with the recipe of FlashSVG-4B or with that of post-training alone (no $\mathcal{L}_{\text{diff}}$ at any stage). The comparisons and the decision rule (within 0.010 DINO of FlashSVG-4B on both kinds: the gain is continued training) were fixed before the runs. With the FlashSVG-4B recipe the control reaches 0.935 / 0.896, a paired difference of 0.000 [−0.014, 0.018] / −0.008 [−0.018, 0.003] (p = 0.97 / 0.14; [−0.018, 0.019] / [−0.024, 0.008] when the three seeds of each arm are resampled as well), and without $\mathcal{L}_{\text{diff}}$ 0.927 / 0.895 (Table S2b). Of the +0.059 / +0.148 that conversion adds, the causal-only updates give +0.050 [0.032, 0.069] / +0.139 [0.114, 0.165], $\mathcal{L}_{\text{diff}}$ in post-training +0.008 / +0.001, and the diffusion objective in place of the causal one 0.000 / +0.008, the last two with intervals that include zero: the diffusion objective provides the drafting mode, not fidelity (we did not measure how well the control, whose diffusion mode is trained only in post-training, drafts). Like the causal term, the causal-only conversion lets text-to-SVG illustration candidates end (0.991 against 0.717), at the judge fidelity of FlashSVG-4B (5.44 / 2.55 against 5.42 / 2.54).

Post-trained with the recipe of FlashSVG-4B (its own round-1 pool, three training seeds), the 4B conversion with $\mathcal{L}_{\text{causal}}$ matches FlashSVG-4B on image-to-SVG DINO (0.935 / 0.904 for both, three seeds each) and termination, and lets text-to-SVG illustration candidates end (0.984 against 0.717). Its judge fidelity is higher by 0.31 / 0.15 (95% CI $[0.07, 0.56]$ / $[-0.01, 0.31]$, p=0.013 / 0.07; per-case scores averaged over the three seeds of each). All other experiments use FlashSVG-4B, converted without the causal term.

S3. Decoding variants and mechanism of short drafts

Exactness and drafts in detail. Along fixed sequences, the decoders' next-token distributions differ by a total variation of at most $6 \cdot 10^{-5}$ in float32 (99th percentile) and 0.005–0.012 on average in bfloat16, with no shift in EOS, and float32 greedy streams match in 158 of 160 cases (the others part at exact ties). One-step drafts match 88% of commands but 52% of coordinates (factorization gap 2.45 against 0.40 nats per token). Whole-vocabulary sampling (Table 1) gives FlashSVG-4B identical samples and moves FlashSVG-8B's DINO by at most 0.002.

Table S2. Conversion of the two released models and the causal term. (a) Causal decoding of conversion checkpoints, one sample on the first 40 MMSVG-Bench image-to-SVG cases per kind (1,536-token cap; these icons are easier than the full set, where the released OmniSVG-4B scores 0.776 instead of 0.837): DINO, share of outputs that end with EOS, and mean generated tokens, icons / illustrations. Conversion follows the recipe of FlashSVG-4B; "+ causal term" adds $\mathcal{L}_{\text{causal}}$ with weight 1, "half learning rate" uses $5 \cdot 10^{-6}$, "+ EOS distillation" adds $\mathcal{L}_{\text{EOS}}$ (Section 4.2) with the given weight, and "output layer frozen" or "at lr $\times 0.1$" keeps the untied 8B output layer at its released values or trains it at a tenth of the learning rate; *this run diverged after update 735. (b) Official protocol of Table 1 (eight candidates, first valid scored): image-to-SVG DINO, share of candidates that end on image-to-SVG and on text-to-SVG, and text-to-SVG judge fidelity, icons / illustrations. Post-trained rows apply the recipe of FlashSVG-4B to their own conversion and round-1 pool, or, without conversion, to the released model and its own pool without $\mathcal{L}_{\text{diff}}$; the causal-only conversion repeats the 4B conversion with $\mathcal{L}_{\text{causal}}$ in place of $\mathcal{L}_{\text{diff}}$ (same data and updates), and its second row is post-trained without $\mathcal{L}_{\text{diff}}$; means of three training seeds.

(a) Conversion checkpoints, causal decoding

Model	DINO↑	Ends↑	Tokens
OmniSVG1.1-8B, released	0.812 / 0.714	0.78 / 0.57	716 / 1,038
conversion, step 500	0.745 / 0.691	0.50 / 0.28	1,012 / 1,231
conversion, step 1,000	0.791 / 0.621	0.60 / 0.22	952 / 1,344
conversion, step 1,500	0.783 / 0.671	0.52 / 0.35	948 / 1,223
conversion, step 2,000	0.790 / 0.648	0.60 / 0.25	876 / 1,297
+ causal term, step 500	0.778 / 0.649	0.50 / 0.25	1,041 / 1,285
+ causal term, step 1,000	0.794 / 0.624	0.60 / 0.28	904 / 1,255
half learning rate, step 500	0.775 / 0.644	0.57 / 0.28	888 / 1,301
half learning rate, step 1,000	0.779 / 0.624	0.45 / 0.20	1,053 / 1,323
half learning rate, step 1,500	0.800 / 0.680	0.55 / 0.30	903 / 1,275
+ EOS distillation, weight 1, step 1,000	0.778 / 0.658	0.60 / 0.32	886 / 1,208
+ EOS distillation, weight 20, step 500	0.803 / 0.676	0.75 / 0.38	661 / 1,078
+ EOS distillation, weight 20, step 1,000	0.857 / 0.663	0.85 / 0.42	642 / 1,125
output layer frozen, step 500	0.766 / 0.620	0.57 / 0.18	982 / 1,336
output layer frozen, step 1,000	0.716 / 0.623	0.48 / 0.18	958 / 1,385
output layer at lr $\times 0.1$, step 500	0.726 / 0.650	0.48 / 0.25	985 / 1,253
output layer at lr $\times 0.1$, step 1,000	0.828 / 0.651	0.50 / 0.32	1,000 / 1,266
vision tower frozen, step 1,000	0.869 / 0.648	0.65 / 0.32	869 / 1,232
OmniSVG-4B, released	0.837 / 0.717	0.70 / 0.48	724 / 992
conversion, step 500	0.847 / 0.648	0.65 / 0.40	784 / 973
conversion, step 1,000	0.813 / 0.707	0.70 / 0.38	616 / 1,069
conversion, step 3,000	0.916 / 0.720	0.78 / 0.42	644 / 1,046
+ causal term, step 500	0.872 / 0.731	0.85 / 0.48	598 / 990
+ causal term, step 3,000	0.886 / 0.793	0.80 / 0.60	603 / 922
+ EOS distillation, weight 20, step 500	0.850 / 0.682	0.72 / 0.48	653 / 916
+ EOS distillation, weight 20, step 1,000*	0.670 / 0.528	0.90 / 0.92	258 / 161
same, second conversion seed, step 1,000	0.820 / 0.641	0.65 / 0.45	761 / 1,012

(b) Official protocol

Model	DINO↑	Ends, image↑	Ends, text↑	Judge↑
OmniSVG-4B + post-training, no conversion	0.877 / 0.756	0.981 / 0.924	0.990 / 0.759	4.84 / 1.97
OmniSVG-4B + conversion	0.816 / 0.728	0.68 / 0.39	–	–
FlashSVG-4B	0.935 / 0.904	0.997 / 0.982	0.979 / 0.717	5.42 / 2.54
4B + causal term, post-trained	0.935 / 0.904	0.995 / 0.984	0.999 / 0.984	5.73 / 2.69
4B causal-only conversion, post-trained	0.935 / 0.896	0.998 / 0.987	0.999 / 0.991	5.44 / 2.55
post-trained without $\mathcal{L}_{\text{diff}}$	0.927 / 0.895	0.998 / 0.990	0.974 / 0.834	5.56 / 2.74
OmniSVG1.1-8B + post-training, no conversion	0.869 / 0.825	0.974 / 0.910	0.943 / 0.699	5.14 / 2.51
OmniSVG1.1-8B + conversion	0.764 / 0.703	0.540 / 0.285	0.703 / 0.303	5.07 / 1.99
8B, conversion with the 4B recipe, post-trained	0.883 / 0.840	0.985 / 0.941	0.962 / 0.758	5.59 / 2.83
FlashSVG-8B (EOS distillation)	0.914 / 0.884	0.981 / 0.925	0.922 / 0.611	5.49 / 2.75

Teacher-forced on the greedy outputs of B (300 cases), a one-step draft of a 16-token block matches the causal greedy choice for its first 6.53 tokens on icons and 5.89 on illustrations, and only 0.089 and 0.059 of blocks match in full. This is greedy acceptance; under sampling, acceptance is lower, and B commits $\tau = 2.40 / 2.27$ tokens per forward pass (Table 2). Command tokens match the causal choice 0.88 of the time, compared with 0.524 for coordinates.

Iterative threshold drafting raises acceptance to 12.45 tokens per block. Its illustration τ is 1.28 at 13.3 s per case, and mean outputs are 534 / 809 tokens. Grammar-constrained drafts, skeleton-then-arguments drafts, and sampled drafts at the generation temperature and at half of it did not improve on one-step greedy drafts (Figure S2). DFlash-style

Table S3. Decoders on fixed weights (MMSVG-Bench image-to-SVG, first valid of eight; icons / illustrations). EOS_{out}: share of scored outputs that terminate. Edit-aware and mask-only checkpoints: trained from B for up to 1,000 steps. Final weights: three sampling runs per decoder (HF-based harness); fused engine (CUDA graphs, fused kernels): one run, timed on the first 40 icons. Bold: best per block.

Decoder	DINO ↑	EOS_{out} ↑	s / icon ↓
Round-1 B weights (λ=0.3)			
Unverified, parallel 4-step	0.360 / 0.311	–	–
Parallel + remask edits	0.433 / 0.361	–	–
Unverified, confidence	0.772 / 0.677	**1.00 / 1.00**	**4.4**
Verified, iterative threshold	**0.899 / 0.840**	0.95 / 0.80	8.3
Edit-aware and mask-only controls			
Parallel + edits, edit-aware	0.303 / 0.301	0.22 / 0.10	32.6
Parallel + edits, mask-only	0.417 / 0.368	–	–
Verified, one-step, edit-aware	0.885 / 0.813	–	–
Verified, one-step, mask-only	**0.904 / 0.837**	–	–
Final weights (λ=30, training seed 1)			
Autoregressive	0.930 / 0.907	0.99 / 0.99	8.5
Verified, one-step (FlashSVG-4B)	0.937 / 0.910	0.99 / 0.99	4.3
Verified, fused engine	**0.938 / 0.911**	0.99 / 0.99	**0.72**

Algorithm 1 Verified self-speculative block-diffusion decoding with one-step greedy drafts.

Require: condition c, weights θ, block length L, cap $T_{\max}$

$x \leftarrow [\,]$
while $|x| < T_{\max}$ **and** $\text{last}(x) \neq \text{EOS}$ **do**
  $\hat{x}_{1:L} \leftarrow \arg\max q_\theta(\cdot \mid x, \text{MASK}^L, c)$ {one diffusion pass}
  $p_{1:L+1} \leftarrow p_\theta(\cdot \mid x, \hat{x}_{1:L}, c)$ {one block-causal pass}
  for $i = 1$ **to** L **do**
    if $u \leq p_i(\hat{x}_i)$ for $u \sim \mathcal{U}(0,1)$ **then**
      append $\hat{x}_i$ to x; **if** $\hat{x}_i = \text{EOS}$ **return** x
    else
      append a sample of p_i without $\hat{x}_i$ (renormalized); **break**
    end if
  end for
  if all L drafts were accepted, append a sample of p_{L+1}
end while
return x

drafter alignment [16] lowers icon DINO (−0.022 [−0.038, −0.006]). The drafter-alignment variants ran for 600 updates each, and the edit-aware arm and mask-only control for up to 1,000 steps from the same checkpoint (Figure S3).

Exactness and token sets. Verified decoding samples from the SVG token set (ids from the first SVG content token on, without BOS after the first token), whereas the release's autoregressive inference samples the whole vocabulary. For FlashSVG-4B the restriction changes nothing (identical samples; training seed 1, one run each: one-sample DINO 0.926 / 0.906, verified 0.938 / 0.918), for FlashSVG-8B little (0.916 / 0.883 on the whole vocabulary, 0.918 / 0.883 on the token set, 0.922 / 0.894 verified; training seed 1, all 150 + 150 cases), and for the released models at most 0.004. The 8B converted with the 4B recipe, however, often skips the BOS token that starts a drawing when it may choose any token (30 of 80 greedy cases), and its whole-vocabulary decoding

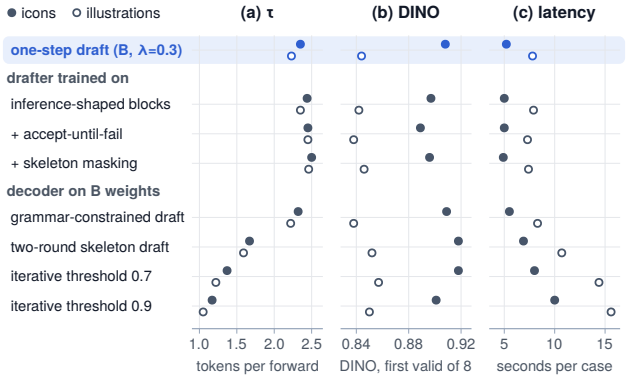


Figure S2. Drafter and decoder variants on the fixed weights of B (λ=0.3): (a) tokens per forward pass, (b) DINO (first valid of eight) and (c) batch-1 seconds per case, icons (filled) and illustrations (open). The shaded row is the one-step draft used throughout.

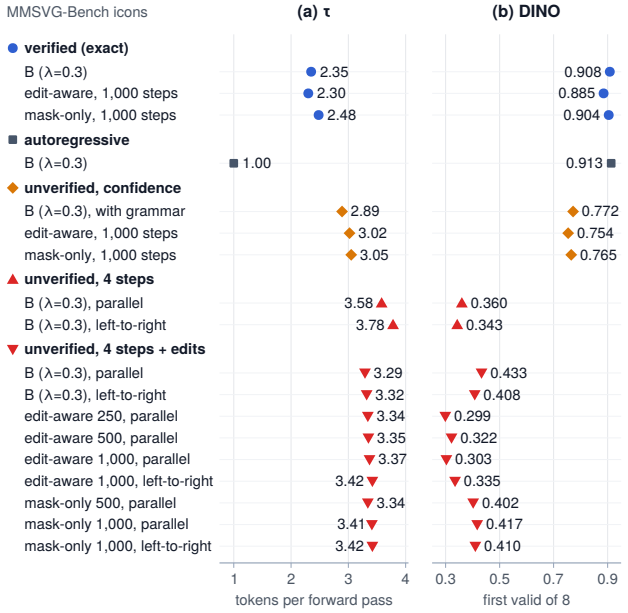


Figure S3. Unverified parallel and edit-aware decoding against verified decoding on MMSVG-Bench icons: (a) tokens per forward pass and (b) DINO (first valid of eight) per decoder and checkpoint (B, λ=0.3, and checkpoints trained edit-aware or mask-only for 250–1,000 steps; confidence decoding commits tokens above probability 0.95). Verified decoding keeps the autoregressive DINO; every unverified decoder loses it, with or without edits.

scores 0.876 / 0.847 against 0.889 / 0.860 on the token set and 0.899 / 0.879 verified. On the token set, greedy verified and autoregressive decoding give identical token streams in float32 on 80 of 80 cases for both 8B models and on 78 of 80 for FlashSVG-4B, whose two exceptions part at exactly tied scores; in bfloat16 the streams part where the two most likely tokens lie within rounding of each other (largest top-2 gap 0.25 and 0.31 logits for the two 8B models, 0.50 for the 4B), after a median of 56–106 tokens. Under sampling, verified decoding scores slightly higher than autoregressive decoding on the token set. A single run per decoder suggested +0.008 / +0.013 pooled over the three post-trained models; with repeated runs (new per-case seeds; three runs per decoder for the two 8B models, five for FlashSVG-4B; per-case means over runs, paired) the differences are +0.004 / +0.010 for the 8B with the 4B recipe, +0.007 / +0.005 for FlashSVG-8B and +0.003 / +0.004 for FlashSVG-4B (no single $p < 0.05$), pooled +0.005 [0.000, 0.009] / +0.005 [0.001, 0.009] (z = 2.1 / 2.3). These pooled runs include the single runs that prompted them; without those, the difference is +0.003 [−0.002, 0.008] / +0.002 [−0.003, 0.007]. With both decoders in float32 (TF32 off, three runs each with the bfloat16 runs' seeds), it is +0.001 [−0.003, 0.006] / −0.002 [−0.007, 0.004]. A token-level comparison locates no source either: along one autoregressive sample per case (first 40 cases per kind, each model; 141,011 positions in float32 and 142,279 in bfloat16), the processed next-token distribution of HF sampling and the distribution the verified decoder draws from (its verifying pass per block, repetition penalty, token set, temperature, top-k and top-p) differ in float32 by a total variation of $1–4 \cdot 10^{-6}$ on average (99th percentile at most $6 \cdot 10^{-5}$; the most likely token never differs) and in bfloat16 by 0.005–0.012 on average (the most likely token differs at 0.5–1.2% of positions), with no token class, EOS included, shifted by more than $3 \cdot 10^{-5}$ in probability; 1,728 chi-square tests of the commit rule (accept the draft with its probability, else resample without it) reject none at Holm-corrected 0.01. The one code difference is the cap: HF counts the forced BOS against it, so autoregressive candidates hold at most 1,535 content tokens and verified ones 1,536, which matters only for the candidates that reach it (about 1% of FlashSVG-4B's). Regenerating a run with the same seeds on another GPU reproduces all 300 scored outputs of each decoder byte for byte.

Batch size. Algorithm 1 decodes one candidate at a time. A batched version decodes the candidates of one case together, with per-row acceptance lengths and a per-row KV-cache offset. Its verify passes agree with a plain causal recomputation as closely as those of the batch-1 decoder (98.65% against 98.64% greedy-token agreement), its candidate lengths match autoregressive decoding at the same batch size (Kolmogorov–Smirnov $p \geq 0.88$), and first-valid

Table S4. Throughput by batch size (candidates of one case decoded together; MMSVG-Bench image-to-SVG, first 40 cases per kind, one H20). Cells are SVG candidates per second, icons / illustrations; the last column is the ratio of output tokens per second, verified over autoregressive decoding of the final weights.

Batch	Released, AR	Ours, AR	Ours, verified	Verified / AR
1	0.07 / 0.04	0.12 / 0.09	**0.26 / 0.17**	1.97× / 1.82×
4	0.17 / 0.15	0.40 / 0.29	**0.74 / 0.45**	1.83× / 1.63×
8	0.33 / 0.29	0.73 / 0.56	**1.26 / 0.83**	1.72× / 1.51×
16	0.59 / 0.56	1.28 / 0.97	**1.73 / 1.11**	1.33× / 1.15×

DINO does not change (all paired 95% intervals include zero). Table S4 reports throughput on one H20 for the first 40 cases per kind. An autoregressive step costs about 20 ms up to batch 16, whereas a verified pass, which processes 16 positions per candidate, rises from 21.5 ms at batch 1 to 30.4 ms at batch 16. The gain over autoregressive decoding of the same weights therefore falls from 1.97× / 1.82× in output tokens per second at batch 1 to 1.72× / 1.51× at batch 8 and 1.33× / 1.15× at batch 16, and at batch 16 it requires removing finished candidates from the batch. Post-training raises candidate throughput at every batch size because outputs are about half as long.

S4. Post-training details

The post-training ablations include target thresholds of 0.8 and 1.0, on-policy rounds 2 and 3, an off-policy round-2 control, and the ground-truth-only arm. Figure S6 shows the rounds study, and Figure S7 shows cases selected by a fixed rule: round-1 arm A did not terminate, whereas the final model did. Figures S8 and S9 show random out-of-domain cases and the non-learned baselines on random cases.

Ending and completeness. The benchmark inputs have no reference SVG, so completeness is measured against the input image: the share of its ink (interior pixels more than 32 levels from its background color) that the scored output, rendered at 224 px, leaves as its own background. FlashSVG-4B leaves 3% / 4% of the ink undrawn on average, the released OmniSVG-4B 21% / 24%; 9% / 13% of FlashSVG-4B's outputs end with more than a tenth undrawn, against 17% / 17% of the release's, which cuts another 25% / 39% at the cap with as much undrawn. At 8B, post-training lowers the undrawn share from 19% / 23% to 6% / 6% (FlashSVG-8B 5% / 5%). The measure flags none of GPT-6 Astra's outputs and 12% / 11% of Claude Opus 5.5's.

The group-relative policy-gradient controls used render rewards, with and without PPO-style ratio clipping. Figure S10 shows their training curves. Figure S11 contrasts the collapse caused by low-score terminating negatives with runaway-only negatives.

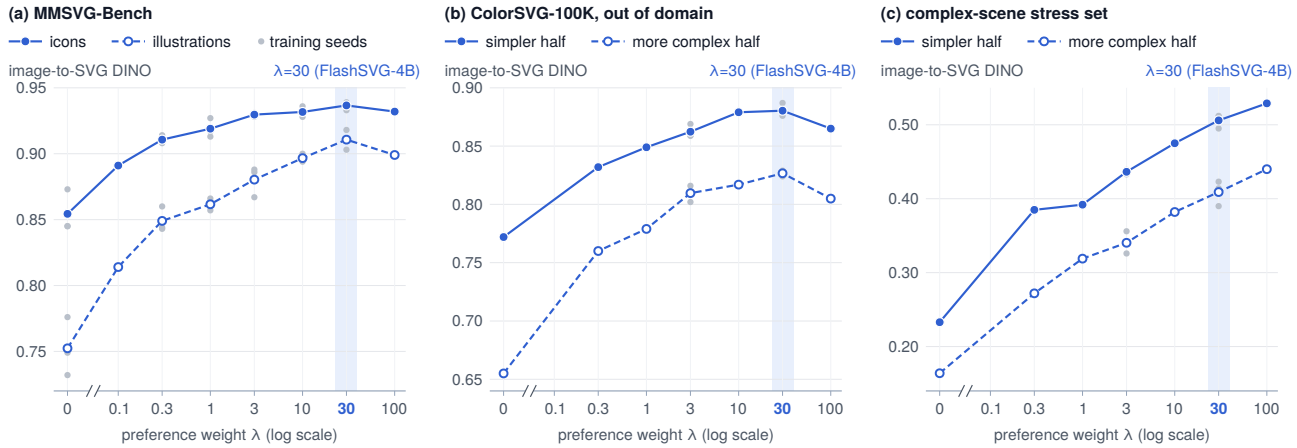


Figure S4. Image-to-SVG DINO (verified decoding, first valid of eight) against the preference weight λ on (a) MMSVG-Bench icons and illustrations and on the simple and complex halves of (b) the out-of-domain ColorSVG-100K and (c) the complex-scene stress set. Lines are means over three training seeds (grey dots) where a weight has three, single runs otherwise; Table 2 lists the same runs except $\lambda = 0.1$ and 1, with their EOS rates and tokens per forward pass. DINO peaks at $\lambda = 30$ (FlashSVG-4B) on MMSVG-Bench and ColorSVG-100K and is still rising at $\lambda = 100$, a single run, on the stress set.

S5. Text-to-SVG details

Direct generation and the cascade. In the cascade, a text-to-image model (FLUX.1-schnell [8]) first produces a picture, which FlashSVG then vectorizes, and a local judge distinct from the GPT-6 Astra evaluator (Qwen3.8-27B [71] rating the same two axes; Spearman correlation with GPT-6 Astra over 636 direct-generation candidates 0.84 / 0.86 on icons and 0.67 / 0.80 on illustrations, fidelity / quality) picks one of 64 vectorized candidates per prompt (the highest sum of its two scores, the better of two selection rules we compared on these prompts); each prompt therefore costs picture generation, 64 FlashSVG decodes, and 64 local-judge evaluations. Over 64 candidates the local-judge pick reaches judge fidelity 8.60 / 7.43 and judge quality 7.31 / 5.78, paired +1.11 [0.82, 1.41] / +1.34 [1.05, 1.63] in fidelity over picking the candidate of one picture by pixel similarity. Both Qwen3.8-27B models remain ahead of it on icons and illustrations (8.87 / 8.77 without and 8.94 / 8.99 with GPT-6 Astra SFT). Direct generation is bounded by the model's own samples: even an oracle GPT-6 Astra pick among the eight direct samples stays below both Qwen3.8-27B models on illustrations. A round of judge-reward reinforcement learning on direct generation yields a modest fidelity gain, with two caveats: selected round-2 illustration outputs terminate in 113/150 cases, against 149/150 after round 1, and 101/150 round-2 illustrations score at most 3. Two attempts to distill cascade outputs into direct generation lowered fidelity.

A second judge. GPT-6 Astra is the text-to-SVG judge, the strongest closed reference, and the teacher of a fine-tuned baseline. Claude Opus 5.5 therefore re-judges the 13 text-to-SVG rows of Table 1 with the same rubric, render and candidates (Table S5; rows with three seeds use training seed 1, compared with GPT-6 Astra on the same seed). It is stricter by about one point but ranks the rows almost identically (Spearman 0.99 on both kinds, Kendall $\tau = 0.94$ / 0.97), agrees per case with Spearman 0.91–0.95 / 0.79–0.83 where both per-case scores are stored, and also places FlashSVG-4B below CTRL-S and both Qwen3.8-27B models. Claude Opus 5.5 also judges its own outputs, which this table marks.

The cascade was run with local-judge selection over 32

Table S5. Text-to-SVG judge fidelity on MMSVG-Bench (0–10, first valid of eight; one sample for API models; icons / illustrations) by GPT-6 Astra (Table 1) and by Claude Opus 5.5 with the same rubric. Claude Opus 5.5 scores GPT-6 Astra's drawings above its own (9.23 / 9.15 against 9.22 / 8.79), so GPT-6 Astra's lead does not rest on its own judge. [1]Training seed 1 for both judges. *Judged by itself.

System	GPT-6 Astra	Claude Opus 5.5
GPT-6 Astra	9.79 / 9.83	9.23 / 9.15
Claude Opus 5.5*	9.71 / 9.44	9.22 / 8.79
Kimi K3	8.83 / 8.57	7.89 / 7.40
Qwen3.5-27B	6.36 / 7.33	4.75 / 5.80
Qwen3.8-27B	8.87 / 8.77	7.82 / 7.71
Qwen3.8-27B + GPT-6 Astra SFT	8.94 / 8.99	8.00 / 8.04
InternSVG-8B	5.67 / 1.89	4.15 / 1.20
CTRL-S (8B)	6.92 / 5.89	5.05 / 3.99
OmniSVG-4B	4.46 / 1.35	3.23 / 0.76
OmniSVG1.1-8B	4.87 / 1.43	3.69 / 0.89
OmniSVG1.1-8B + post-training[1]	5.33 / 2.44	4.09 / 1.63
FlashSVG-8B[1]	5.49 / 2.84	4.21 / 1.68
FlashSVG-4B[1]	5.43 / 2.55	4.09 / 1.59

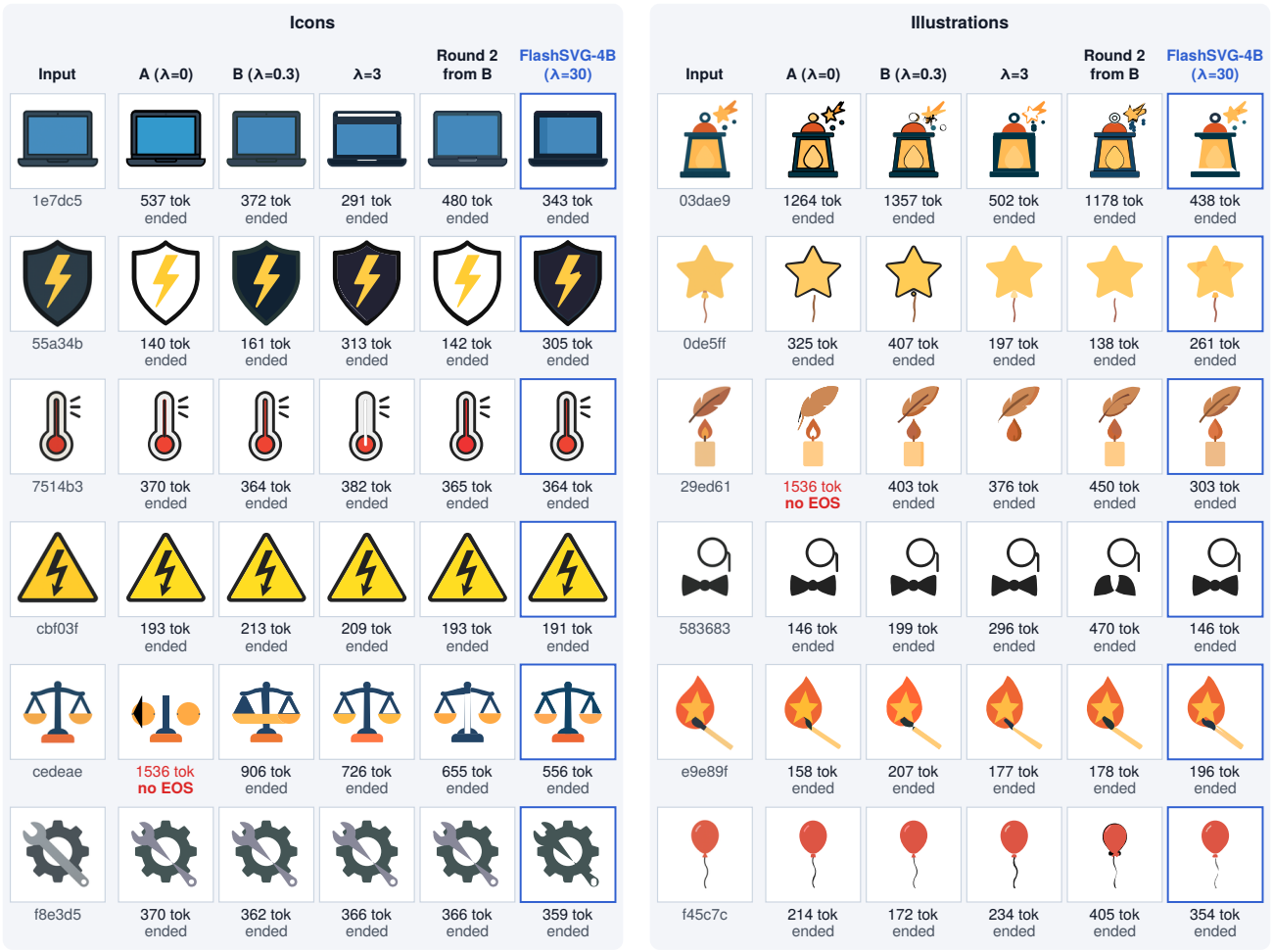


Figure S5. Random MMSVG-Bench image-to-SVG cases: six icons (left) and six illustrations (right) drawn with a fixed seed (20261006) among the 150 cases per kind scored by every run, in id order. Columns: input; round-1 arms A (λ=0, targets only) and B (λ=0.3); λ=3; a second on-policy round from B; and FlashSVG-4B (λ=30). Each cell renders the run's scored output (first valid of up to eight candidates, verified decoding), with the number of generated SVG tokens and whether the output ended with EOS below it (red: no EOS, cut at the 1,536-token cap); the six-character case id is under each input.

and over 64 candidates. Figure S13 shows the 32-candidate judge picks, and Figures S14 and S15 show direct text-to-SVG outputs.

Round 3 of judge-reward reinforcement learning is reported as provisional. Two distillation runs that trained direct generation on cascade outputs lowered fidelity. A vectorizer post-trained on the picture domain was also evaluated inside the cascade. Figure S16 tracks text-to-SVG fidelity and the paired differences across these variants. Figure S17 explains why DINO under-ranks DiffVG.

S6. Further baselines, transfers, and engines

Further learned baselines. Table S8 adds the learned SVG generators of the related-work audit, each with its released prompt, sampling, token budget, and decoder under the official protocol (eight candidates, first valid, invalid scored as 0). CTRL-S is the strongest of them on image-to-SVG and terminates every candidate; on text-to-SVG it is far ahead of FlashSVG, as are the Qwen3.8-27B models (Table S6). StarVector-1B is strong on icons but rarely finishes illustrations. HiVG-3B is evaluated with its only public checkpoint, HiVG-3B-Base, through the authors' inference code and release settings (their prompts, temperature 0.7, top-p 0.9, top-k 50, no repetition penalty, 4,096 new tokens), on inputs prepared at 448 px on white as for every model; outputs without a closing `</svg>` count as invalid, as in the release's own post-processing. Its configuration indicates a 224-pixel stage-2 model rather than the final model of the HiVG paper, and segment tokens are dropped because their table is unreleased. Most of its image-to-SVG samples

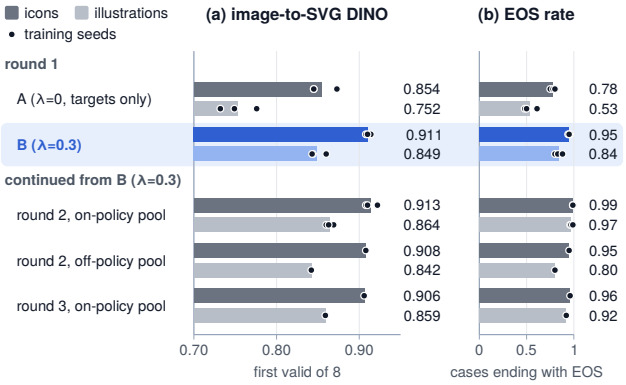


Figure S6. Post-training rounds on MMSVG-Bench image-to-SVG: (a) DINO, first valid of eight; (b) EOS rate, the share of the 150 cases per kind whose scored output ends with EOS. Bars are seed means and dots single training seeds (three for A (λ=0, targets only), B (λ=0.3) and on-policy round 2; one for the other arms). An on-policy round 2 from B helps only where termination is not yet fixed: on illustrations it raises the EOS rate from 0.84 to 0.97 and DINO from 0.849 to 0.864, whereas icons, already at an EOS rate of 0.95, stay at 0.911–0.913. Round 2 on the off-policy round-1 pool and a third round add nothing.

Table S6. Text-to-SVG judge fidelity (GPT-6 Astra judge, 0–10, first valid) on MMSVG-Bench, with the number of candidates per prompt (Cand.). Bold marks the best open value per column; the GPT-6 Astra rows use the closed model. FlashSVG-4B rows use training seed 1.

System	Icons ↑	Illus. ↑	Cand.
Closed reference			
GPT-6 Astra (low effort)	9.79	9.83	1
Open models			
OmniSVG-4B [102]	4.46	1.35	8
Qwen3.8-27B [71]	8.87	8.77	8
Qwen3.8-27B + GPT-6 Astra SFT	**8.94**	**8.99**	8
Ours			
R1-B (λ=0.3), direct	4.80	1.85	8
FlashSVG-4B, direct	5.43	2.55	8
FlashSVG-4B, GPT-6 Astra pick of 8	7.38	4.05	8
Cascade, judge pick of 64	8.60	7.43	64

repeat until the token limit (17% / 7% of candidates end), and an earlier run with our own runner gave the same scores (DINO 0.294 / 0.142); the HiVG paper's numbers come from a different test set and model and are not comparable. VinciCoder-7B is timed with vLLM only; StarVector-1B and 8B and HiVG-3B are timed with their release decoders, one sample per request, on the first 10 cases per kind. Figures S23 and S24 show the first cases by id, unselected.

API-served baselines. The eight API models of Table 1 draw one sample per case with a 24,000-token cap and the lowest reasoning setting each endpoint accepts (low for GPT-6 Astra and GLM-5.3-Flash, minimal for Claude Opus 5.5 and Fable 5.1, none otherwise); requests run 8–16 at a time and back off on rate limits, and their seconds are the wall-clock time of the answering request. With a 200-unit canvas in the image-to-SVG prompt, Claude Opus 5, Opus 4.8 and Kimi K3 often drew in the input's 448-pixel coordinates, so that more than a tenth of the drawing fell outside the canvas in 61%, 38% and 69% of their replies (GLM-5.3-Flash 9%, the others none), cropping it (Figure S18). Asking for the input's pixel grid (viewBox 0 0 448 448) removes the overflow for every model and raises their DINO from 0.757 / 0.638, 0.776 / 0.739 and 0.634 / 0.521 to 0.915 / 0.890, 0.885 / 0.856 and 0.922 / 0.895; Table 1 reports this canvas for every API model (Claude Fable 5.1, which never overflowed, scores 0.930 / 0.908 on the 200-unit canvas). Text-to-SVG has no input image and keeps its prompt. A quarter of the DeepSeek-V4.1-Flash requests were served by an upstream that reports only "deepseek-flash"; their scores match the rest (DINO 0.893 / 0.842 against 0.875 / 0.846).

Rate–distortion against vtracer. vtracer at its defaults beats every learned model on MMSVG-Bench image-to-SVG with files about eight times larger than FlashSVG's (Table 1), so we sweep 312 simplifying settings on the same 448-pixel inputs and score each with the scorer of Table 1 (Figure S19): region settings (speckle filter 4–64, color precision 2–6, layer difference 16–64, path precision) and curve settings (tracing at 448, 224 or 112 pixels, length threshold 4 or 10, splice threshold 45 or 90, corner threshold 60 or 120). FlashSVG-4B's scored outputs average 3.16 / 4.34 KB, 5.6 / 9.2 paths and 119 / 155 path commands (icons / illustrations, three seeds). The best setting whose mean size does not exceed FlashSVG's reaches DINO 0.951 / 0.960 (FlashSVG 0.935 / 0.904); at the same number of path commands 0.935 / 0.954, and at the same gzip-compressed size, where FlashSVG's compact path-token output helps, 0.908 / 0.907. Capping each case at FlashSVG's size for that case, which no single setting meets, and taking the best setting per case by DINO itself, vtracer reaches 0.900 / 0.940 against 0.941 / 0.904 on the cases where a setting fits. On these clean renders, tracing therefore matches or beats FlashSVG at equal size except under the per-case cap on icons; the learned model's advantages are conditioning on text and a single model for both tasks, not compression.

Engine-matched latency. Batch-1 latency depends on the inference engine as much as on the model (Table S9). The FlashSVG export is a standard vision-language checkpoint, so its autoregressive mode runs on vLLM unchanged; on all 600 MMSVG-Bench cases this gives DINO 0.935 / 0.907 and CLIP 0.276 / 0.225, against 0.936 / 0.900 and 0.275 / 0.223 for our harness (training seed 1 in both). On vLLM



Figure S7. Image-to-SVG cases selected by a rule: the scored output of round-1 arm A (λ=0) did not end with EOS, whereas that of FlashSVG-4B (λ=30) did. The rule holds for 37 of the 150 MMSVG-Bench icons and 77 of the 150 illustrations; the first five of each by id are shown (icons left, illustrations right). Columns as in Figure S5. Each cell renders the scored output (first valid of up to eight candidates, verified decoding), with its generated SVG tokens and EOS status below (red: no EOS, cut at the 1,536-token cap).

every model decodes at 142–162 tokens per second at batch 1, and FlashSVG is the fastest because post-training shortens its programs (390 / 581 tokens per image-to-SVG sample against 622 / 860 for CTRL-S). vLLM's n-gram speculation raises throughput by 8% once the repetition penalty of the official sampling is applied to drafts. Verified block-diffusion decoding exists only in our harness, where it halves the forward passes of the same weights; under the same harness CTRL-S takes 15.1 / 20.1 s per case. On vLLM, FlashSVG's autoregressive readout (2.5 / 3.6 s) is faster than verified decoding in our HF-based harness (4.3 / 6.0 s), so speed comparisons across rows must be engine-matched. Removing the harness's per-step overhead reverses this. With CUDA graphs, fused kernels and a batch-1 attention kernel, the same weights decode autoregressively in 1.43 / 2.02 s and with verification in 0.72 / 1.10 s per case (0.84 / 1.17 s on a second sampling pass), against 2.90 / 3.98 s for the first sample of the released OmniSVG-4B on the same engine (apportioned by forward passes; 3.07 / 5.23 s to its first valid candidate). Verified decoding is thus 2.9–3.4× faster than vLLM, whose timing is unchanged with its default multiprocessing engine (2.51 / 3.69 s). A static-shape tier that replays the harness's own kernels reproduces verified decoding token for token. The fused tier changes the bf16 arithmetic, so greedy outputs part after a median of 52 tokens, but lengths, termination and paired DINO match (all 95% intervals include zero), and on all 600 cases it gives DINO 0.938 / 0.911 and CLIP 0.276 / 0.223.

Prompt-lookup drafts. Training-free prompt-lookup drafts (longest earlier match of three to one tokens, up to 16 tokens) inside the same verifier commit 1.09 / 1.13 tokens per forward, so they run at autoregressive speed (8.15 / 11.34 s per case, against 4.06 / 5.95 s with diffusion drafts in the same run). A hybrid that uses a lookup draft when the match has at least two tokens ties verified decoding. SVG coordinates rarely repeat exactly: a lookup draft is accepted for 0.15 tokens after a one-token match and 1.7–2.0 tokens after a three-token match. All decoders sample exactly (identical greedy tokens on 16 of 16 cases; DINO within 0.015).



Figure S8. Random out-of-domain image-to-SVG cases from ColorSVG-100K (300 test icons split at the median number of drawing elements): six from the simpler half (left) and six from the more complex half (right), drawn with a fixed seed (20261006), in id order. Columns: input; round-1 arms A (λ=0) and B (λ=0.3); λ=3; FlashSVG-4B (λ=30). Each cell renders the scored output (first valid of up to eight candidates, verified decoding), with its generated SVG tokens and EOS status below (red: no EOS, cut at the 1,536-token cap).

Transfer to other model families. One round of termination-aware post-training transfers to model families with raw SVG code (Table S10). On StarVector-8B, every preference weight from 1 to 30 raises DINO and termination on both benchmarks, while weight 0 on the same targets does not (Figures S25 and S26). On Qwen3.8-27B, termination rises with LoRA and with full-parameter training alike, and text-to-SVG judge fidelity is unchanged (8.81 / 8.70 before and 8.89 / 8.66 after on SArena-600). The transfers to other families use LoRA (rank 32, vision encoder frozen) unless marked full-parameter; the full-parameter runs use the main recipe's optimizer settings and the same data and updates. With 69 updates, full-parameter training of StarVector-8B at peak learning rates of 5×10^{-6}, 10^{-5}, 2×10^{-5} and 5×10^{-5} gives MMSVG-Bench DINO 0.839 / 0.697, 0.812 / 0.658, 0.763 / 0.593 and 0.817 / 0.705 (candidates that end 0.73 / 0.54, 0.70 / 0.54, 0.64 / 0.47 and 0.85 / 0.72): it matches LoRA on icons but fixes illustrations only in part. The StarVector-8B row at λ=10 is the mean of three training seeds (MMSVG-Bench DINO 0.832–0.841 / 0.772–0.824, candidates that end 0.93–0.95 / 0.86–0.92; SArena-600 DINO 0.839–0.875 / 0.446–0.508, candidates that end 0.95–0.97 / 0.45–0.61). The recipe needs targets in the model's own style: VinciCoder-7B's own SVGs rarely

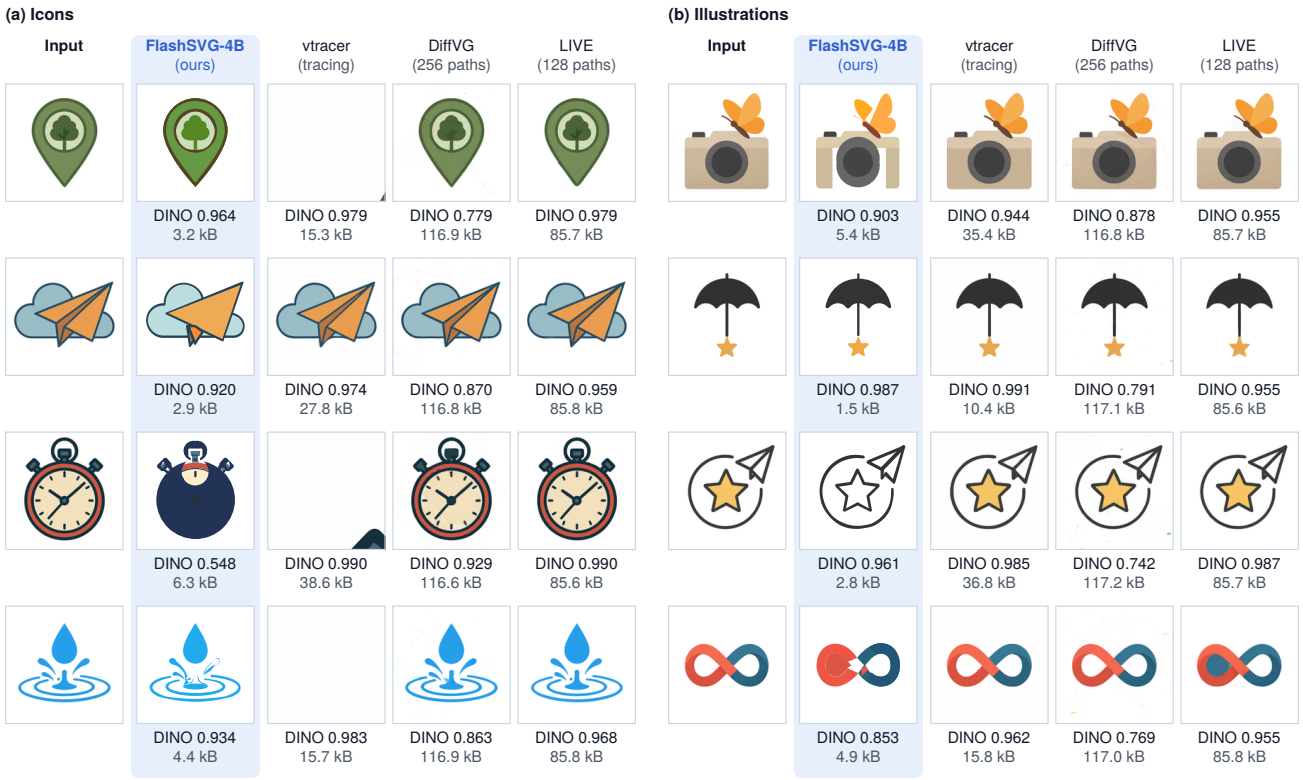


Figure S9. Image-to-SVG against non-learned vectorizers on MMSVG-Bench: (a) four icons and (b) four illustrations drawn at random (seed 7, unselected) among the 299 of 300 cases where every method returned an SVG (FlashSVG-4B has no valid candidate on the remaining icon). Columns: input; FlashSVG-4B (first valid of eight candidates); vtracer; DiffVG with 256 paths; LIVE with 128 paths. Under each output: its DINO similarity to the input and its SVG file size. DINO under-ranks DiffVG (Figure S17).

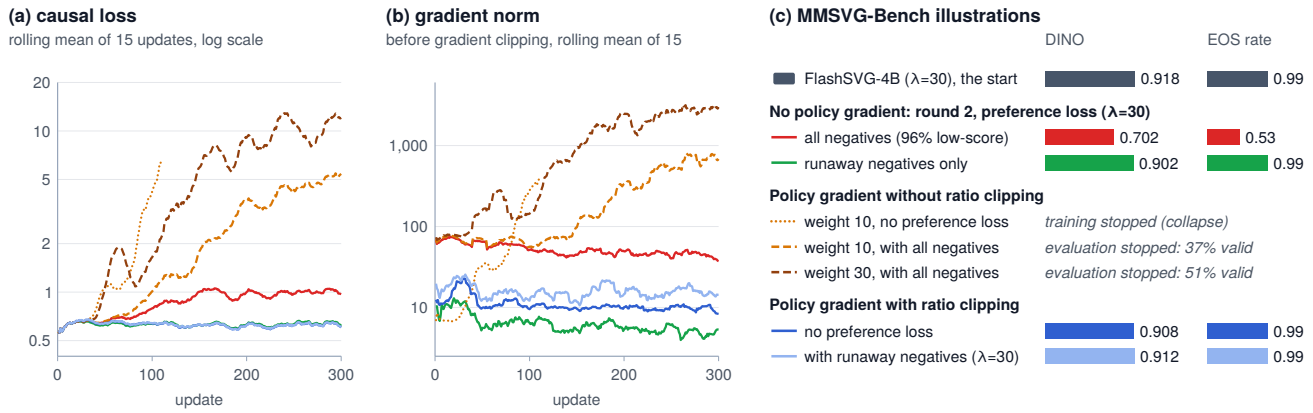


Figure S10. Second-round training from FlashSVG-4B (λ=30): (a) causal loss on the targets and (b) gradient norm before clipping (rolling means of 15 updates) for round-2 preference training with all or runaway-only negatives and for render-reward group-relative policy optimization with and without ratio clipping; (c) MMSVG-Bench illustration DINO and EOS rate of each evaluated run. Without ratio clipping the policy-gradient runs diverge; with it they stay at the start, and low-score negatives collapse termination.

reach the fidelity threshold, so 97% of its targets were ground truth in another style, and the model became worse; with its best terminating candidates as targets instead (a variant of the target rule), every candidate ends, outputs are 1.6× shorter, DINO is unchanged on MMSVG-Bench, and SArena-600 illustrations improve (Figures S27 and S28).



Figure S11. Second-round training from FlashSVG-4B on image-to-SVG cases selected by a rule: FlashSVG-4B (λ=30, the starting point) ended its output with EOS and the second preference round with negatives as in round one did not. The rule holds for 40 of the 150 MMSVG-Bench icons and 69 of the 150 illustrations; the first five of each by id are shown (icons left, illustrations right). Columns: input; FlashSVG-4B; round 2 with negatives as in round one (96% low-score) or with runaway negatives only; render-reward GRPO with PPO-style ratio clipping, without a preference loss and with the preference loss on runaway negatives. Each cell renders the scored output (first valid of up to eight candidates, verified decoding), with its generated SVG tokens and EOS status below (red: no EOS, cut at the 1,536-token cap).

Decoding controls. Decoding-time rules do not substitute for post-training (Table S11). Banning exactly repeated 8-token sequences raises the termination of the released OmniSVG-4B but lowers DINO and judge fidelity significantly, because SVG code repeats short sequences legitimately; longer bans and the presence penalty that Qwen3.8-27B's model card recommends against endless repetition change little.

SArena-600 text-to-SVG. Figure S29 compares judge fidelity on the second benchmark. The cascade (a text-to-image picture vectorized by FlashSVG, pixel pick among eight) reaches 7.24 / 6.22 at about 21 / 33 GPU-seconds per prompt, and a local-judge pick over four pictures reaches 8.27 / 6.89. CTRL-S reaches 7.74 / 6.13 and direct FlashSVG 5.92 / 2.90 (training seed 1, whose outputs the cascade vectorizes); Qwen3.8-27B stays highest.

Table S7. All systems of Table 1, with the rows left out of the main table. MMSVG-Bench [102] under the official protocol (up to eight candidates; first valid scored; invalid scored as 0). Image-to-SVG: DINO, SSIM and LPIPS [112] against the input (invalid outputs count as SSIM 0 and LPIPS 1), EOS = fraction of candidates that terminate, s = batch-1 seconds per case on one H20 (open models with HF generate; FlashSVG: verified decoding in our HF-based harness; other engines in Supplementary Table S9). FlashSVG candidates are sampled autoregressively from its weights, which verified decoding samples exactly. Text-to-SVG: CLIP and judge fidelity (GPT-6 Astra judge, 0–10). API models: one sample, lowest reasoning setting (low for GPT-6 Astra), 24k-token cap; API and general vision-language models draw on the input's 448-pixel canvas. Baselines are single evaluations; the rows of FlashSVG-4B, FlashSVG-8B and OmniSVG1.1-8B + post-training are means of three training seeds. Bold marks the best open-weight learned value per column (not for s, whose engines differ); closed models are references. "–": not measured or not applicable. [a]Remote API wall-clock, not comparable. [s]Timed on one training seed. [d]Fine-tuned by us on GPT-6 Astra SVGs. [v]Timed with vLLM. [h]The only public checkpoint, HiVG-3B-Base, run with the authors' inference code and settings (Supplementary Section S6). [c]CPU. [e]Conversion with EOS distillation (Section 4.2). [†]DINO under-ranks DiffVG, whose faint stray strokes cost DINO but hardly SSIM or LPIPS (Supplementary Figure S17).

	Icons							Illustrations						
System	**DINO↑**	**SSIM↑**	**LPIPS↓**	**CLIP↑**	**Judge↑**	**EOS↑**	**s↓**	**DINO↑**	**SSIM↑**	**LPIPS↓**	**CLIP↑**	**Judge↑**	**EOS↑**	**s↓**
Closed models (API)														
GPT-6 Astra [64]	0.970	0.972	0.021	0.308	9.79	–	28.3[a]	0.981	0.984	0.009	0.313	9.83	–	35.1[a]
Claude Opus 5.5 [4]	0.960	0.890	0.057	0.304	9.71	–	9.7[a]	0.939	0.900	0.061	0.316	9.44	–	12.7[a]
Claude Opus 5 [3]	0.915	0.810	0.135	0.307	9.39	–	10.4[a]	0.890	0.833	0.136	0.309	9.33	–	13.6[a]
Claude Opus 4.8 [2]	0.885	0.783	0.183	0.307	9.21	–	8.6[a]	0.856	0.808	0.183	0.311	8.98	–	10.7[a]
Claude Fable 5.1 [1]	0.942	0.846	0.092	0.306	9.51	–	8.0[a]	0.914	0.855	0.103	0.314	9.37	–	10.1[a]
Open-weight general models (API)														
Kimi K3 [43]	0.922	0.880	0.092	0.304	8.83	–	24.6[a]	0.895	0.872	0.109	0.306	8.57	–	27.6[a]
DeepSeek-V4.1-Flash [23]	0.878	0.784	0.196	0.304	8.71	–	7.0[a]	0.845	0.809	0.196	**0.311**	8.97	–	8.0[a]
GLM-5.3-Flash [108]	0.901	0.795	0.166	**0.307**	**9.09**	–	21.8[a]	0.883	0.821	0.156	0.309	8.70	–	25.9[a]
Non-learned image-to-SVG														
vtracer [69]	0.982	0.987	0.009	–	–	–	0.02[c]	0.978	0.987	0.009	–	–	–	0.02[c]
DiffVG [48]	0.776[†]	0.986	0.023	–	–	–	70	0.797[†]	0.984	0.030	–	–	–	64
LIVE [59]	0.945	0.994	0.006	–	–	–	792	0.955	0.994	0.006	–	–	–	779
AdaVec [113]	0.935	0.979	0.024	–	–	–	83	0.947	0.986	0.018	–	–	–	80
Open general vision-language models														
Qwen3.5-4B [70]	0.695	0.652	0.345	0.270	4.06	0.51	84.5	0.475	0.468	0.549	0.260	4.49	0.29	94.2
Qwen3.5-9B [70]	0.473	0.443	0.578	0.274	4.83	0.40	101.8	0.289	0.294	0.721	0.273	5.83	0.23	94.6
Qwen3.5-27B [70]	0.923	0.873	0.093	0.289	6.36	0.87	99.1	0.833	0.812	0.172	0.294	7.33	0.72	110.6
Qwen3.8-27B [71]	0.887	0.804	0.171	**0.307**	8.87	0.87	107.6	0.809	0.763	0.226	0.306	8.77	0.69	135.1
Open SVG models														
Qwen3.8-27B + GPT-6 Astra SFT[d]	0.889	0.767	0.226	0.303	8.94	0.97	67.5	0.872	0.777	0.245	0.298	**8.99**	0.96	77.2
InternSVG-8B [87]	0.863	0.819	0.161	0.282	5.67	0.81	–	0.840	0.842	0.155	0.157	1.89	0.59	–
CTRL-S (8B) [88]	0.899	0.822	0.142	0.295	6.92	**1.00**	15.1	0.854	0.836	0.159	0.276	5.89	**1.00**	20.1
StarVector-1B [75]	0.896	0.812	0.143	–	–	0.77	15.7	0.698	0.669	0.315	–	–	0.46	27.2
StarVector-8B [75]	0.729	0.679	0.314	–	–	0.47	36.5	0.520	0.502	0.500	–	–	0.23	53.6
VinciCoder-7B [114]	0.781	0.654	0.505	–	–	0.92	12.6[v]	0.746	0.698	0.499	–	–	0.78	30.2[v]
HiVG-3B[h] [100]	0.283	0.366	0.752	0.253	4.25	0.17	78.0	0.151	0.220	0.872	0.108	1.27	0.07	94.0
OmniSVG-4B [102]	0.776	0.837	0.197	0.271	4.46	0.63	17.3	0.707	0.834	0.245	0.215	1.35	0.40	26.9
OmniSVG1.1-8B [102]	0.800	0.848	0.188	0.270	4.87	0.71	12.8	0.690	0.831	0.262	0.218	1.43	0.45	24.6
Ours														
OmniSVG1.1-8B + post-training	0.869	0.885	0.122	0.270	5.14	0.97	8.2[s]	0.825	0.882	0.148	0.223	2.51	0.91	11.3[s]
FlashSVG-8B[e] (λ=30)	0.914	0.909	0.088	0.274	5.49	0.981	4.5	0.884	0.910	0.101	0.229	2.75	0.925	6.8
OmniSVG-4B + block-diff. conversion	0.816	0.862	0.159	0.274	–	0.68	–	0.728	0.862	0.211	0.214	–	0.39	–
+ post-train., λ=0.3 (R1-B)	0.908	0.910	0.087	0.270	4.80	0.97	5.2	0.844	0.888	0.133	0.216	1.85	0.82	7.8
FlashSVG-4B (λ=30)	**0.935**	**0.928**	**0.064**	0.276	5.42	0.997	4.3	**0.904**	**0.918**	**0.083**	0.223	2.54	0.982	6.1

Table S8. Further learned baselines on MMSVG-Bench [102] and SArena-600 [87] (eight candidates, first valid; icons / illustrations). Image-to-SVG: DINO and the fraction of candidates that end; text-to-SVG: judge fidelity (GPT-6 Astra judge, 0–10). "–" marks a task the model does not perform. Our rows are means of three training seeds. DINO^{-}: SArena-600 without the 45 image inputs (14 icons, 31 illustrations) that have a near-duplicate among our training renders (Supplementary Section S1); the MMSVG-Bench inputs have none.

	MMSVG-Bench			SArena-600			
System	**DINO ↑**	**Cand. EOS ↑**	**Judge ↑**	**DINO ↑**	**DINO^{-} ↑**	**Cand. EOS ↑**	**Judge ↑**
Image- and text-to-SVG							
UniSVG-7B [47]	0.737 / 0.676	0.80 / 0.63	5.03 / 1.35	0.752 / 0.363	0.752 / 0.290	0.87 / 0.28	6.53 / 3.10
InternSVG-8B [87]	0.863 / 0.840	0.81 / 0.59	5.67 / 1.89	0.922 / 0.674	0.921 / 0.637	0.92 / 0.45	6.82 / 3.80
HiVG-3B^{h} [100]	0.283 / 0.151	0.17 / 0.07	4.25 / 1.27	0.434 / 0.094	0.447 / 0.074	0.39 / 0.06	6.28 / 2.27
CTRL-S (8B) [88]	0.899 / 0.854	1.00 / 1.00	6.92 / 5.89	0.904 / 0.745	0.901 / 0.723	1.00 / 0.99	7.74 / 6.13
Image-to-SVG							
StarVector-8B [75]	0.729 / 0.520	0.47 / 0.23	–	0.797 / 0.125	0.803 / 0.073	0.74 / 0.06	–
StarVector-1B [75]	0.896 / 0.698	0.77 / 0.46	–	0.867 / 0.209	0.874 / 0.118	0.80 / 0.12	–
VinciCoder-7B [114]	0.781 / 0.746	0.92 / 0.78	–	0.802 / 0.444	0.800 / 0.379	0.90 / 0.35	–
RefineSVG-7B [54]	0.810 / 0.749	1.00 / 1.00	–	0.832 / 0.649	0.832 / 0.629	1.00 / 1.00	–
Text-to-SVG							
IconShop [95]	–	–	4.39 / 1.90	–	–	–	2.55 / 0.96
vHector-8B [116]	–	–	3.34 / 0.85	–	–	–	3.62 / 0.98
SVGen-7B [85]	–	–	5.05 / 2.06	–	–	–	4.78 / 1.83
IntroSVG-7B [86]	–	–	4.45 / 2.39	–	–	–	5.15 / 2.63
SVGThinker-7B [15]	–	–	4.82 / 3.07	–	–	–	6.24 / 3.88
FlashSVG-4B (λ=30)	0.935 / 0.904	0.997 / 0.982	5.42 / 2.54	0.922 / 0.767	0.922 / 0.753	0.994 / 0.661	6.12 / 2.87
FlashSVG-8B (λ=30)	0.914 / 0.884	0.981 / 0.925	5.49 / 2.75	0.887 / 0.726	0.883 / 0.698	0.966 / 0.461	6.00 / 2.86



Figure S12. MMSVG-Bench text-to-SVG, selected cases: three icons and three illustrations picked by us for visual appeal among the prompts on which FlashSVG-4B reaches GPT-6 Astra judge fidelity ≥ 8 (icons) or ≥ 6 (illustrations) and GPT-6 Astra, Claude Opus 5.5 and Qwen3.8-27B all return a valid SVG (44 and 8 of 150); unselected cases in Supplementary Figure S22. Each tile is the scored output (first valid of eight; one sample for API models; grey: no valid output). On most prompts FlashSVG-4B trails Qwen3.8-27B and the API models (Table 1).

Table S9. Batch-1 seconds per case on one H20 (one sample per request; first 40 cases per task and kind, except in our HF-based harness: all 150 per kind, training seed 1, three runs; icons / illustrations), grouped by inference engine. On our fused engine, seconds run to the first valid output, as in the main text (the released model's first image-to-SVG sample: 2.9 / 4.0 s, apportioned by forward passes).

System	Image-to-SVG ↓	Text-to-SVG ↓
vLLM [45]		
OmniSVG-4B [102], AR	5.3 / 7.2	3.5 / 7.0
CTRL-S (8B) [88]	4.5 / 6.1	3.3 / 6.0
RefineSVG-7B [54]	5.4 / 7.5	–
VinciCoder-7B [114]	12.6 / 30.2	–
UniSVG-7B [47]	19.1 / 27.3	10.8 / 14.4
FlashSVG-4B, AR + n-gram	2.5 / 3.4	2.3 / 5.9
FlashSVG-4B, AR	2.5 / 3.6	3.5 / 6.4
HF generate (CTRL-S) and our harness (FlashSVG)		
CTRL-S (8B) [88]	15.1 / 20.1	12.2 / 19.5
FlashSVG-4B, AR	8.5 / 11.1	–
FlashSVG-4B, verified	4.3 / 6.0	–
Our fused engine (CUDA graphs, fused kernels)		
OmniSVG-4B [102], AR	3.1 / 5.2	2.3 / 6.0
FlashSVG-4B, AR	1.4 / 2.0	1.4 / 3.4
FlashSVG-4B, verified	**0.7 / 1.1**	**1.0 / 3.3**

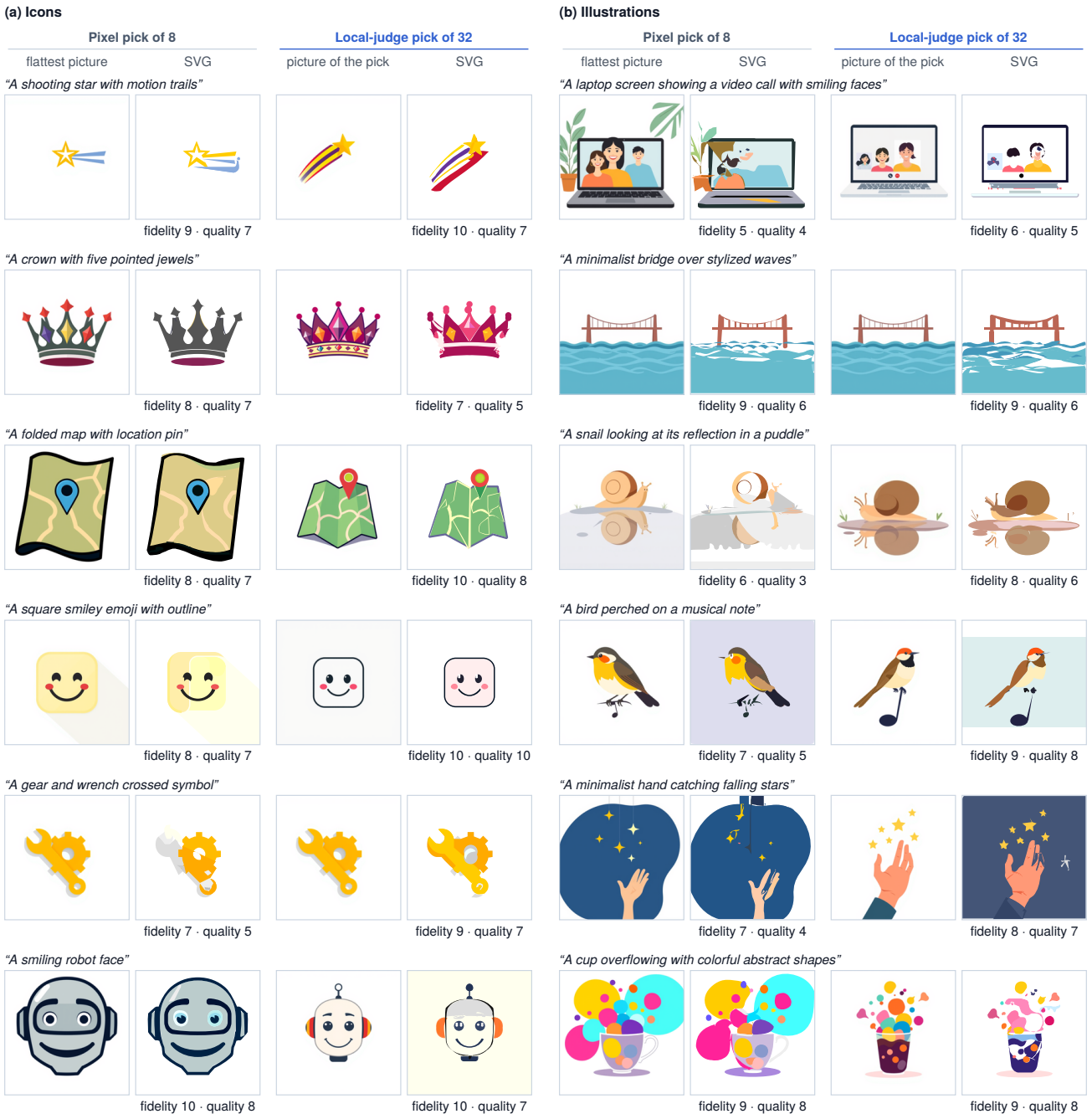


Figure S13. Text-to-SVG cascade on (a) six icon and (b) six illustration prompts of MMSVG-Bench drawn at random (seed 20261006, unselected). Each prompt has four text-to-image pictures, each vectorized eight times by FlashSVG-4B. Pixel pick of 8: the flattest picture (fewest colours, chosen without the prompt) and, of its eight vectorizations, the one closest to it in pixels. Local-judge pick of 32: the vectorization the local judge ranks first among all 32, with the picture it came from. Under each SVG: GPT-6 Astra judge fidelity and quality (0–10). The cascade row of Table S6 picks among 64 candidates of eight pictures instead.

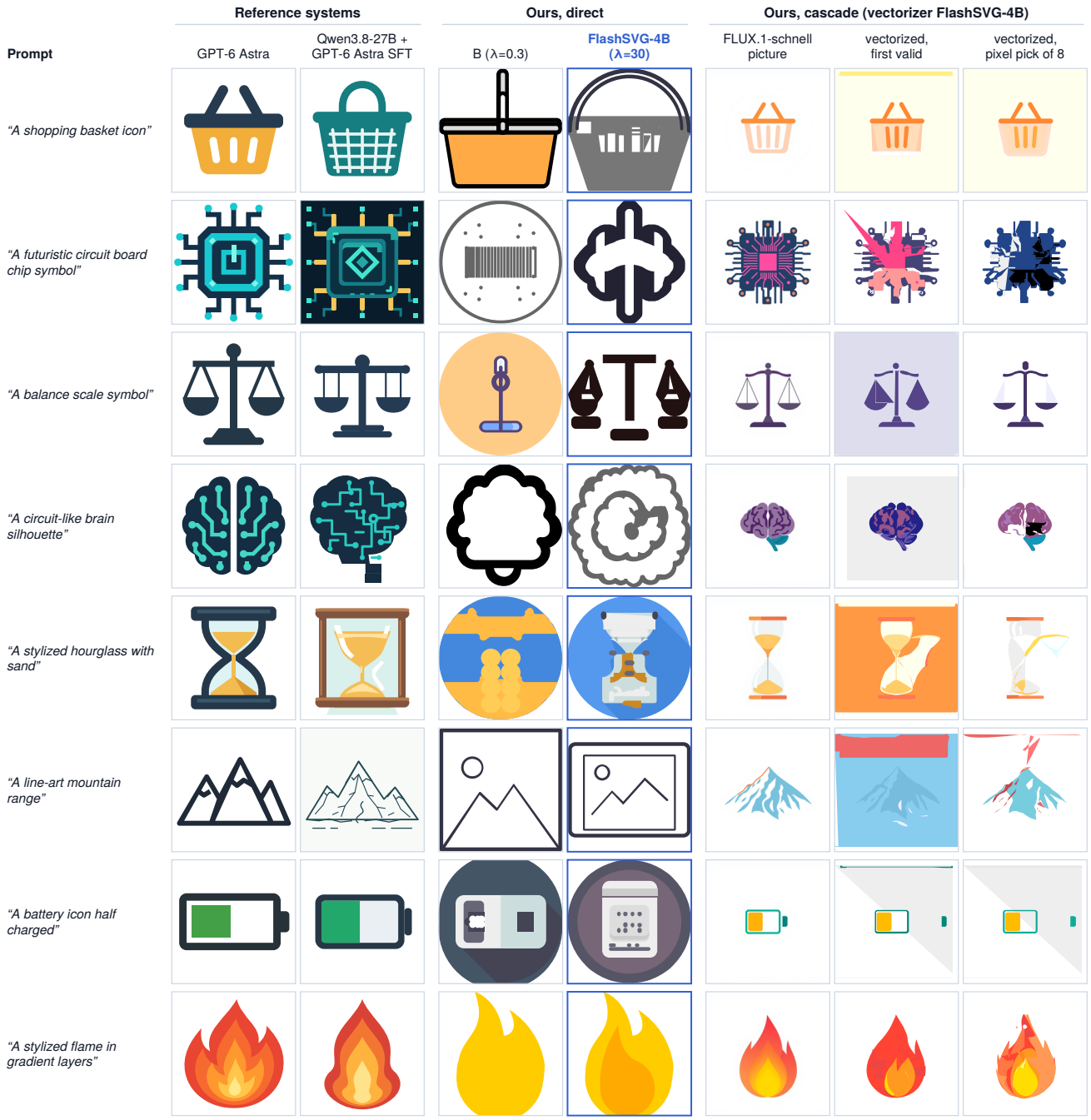


Figure S14. MMSVG-Bench text-to-SVG icons, the first eight prompts by id (unselected). Reference systems: GPT-6 Astra and Qwen3.8-27B + GPT-6 Astra SFT (one sample per prompt). Ours, direct: B (λ=0.3) and FlashSVG-4B (λ=30), first valid of eight candidates. Ours, cascade: a FLUX.1-schnell picture (flat-style prompt, flattest of four draws), its vectorization by FlashSVG-4B (first valid of up to eight candidates), and the one of eight vectorizations picked by pixel similarity to the picture.

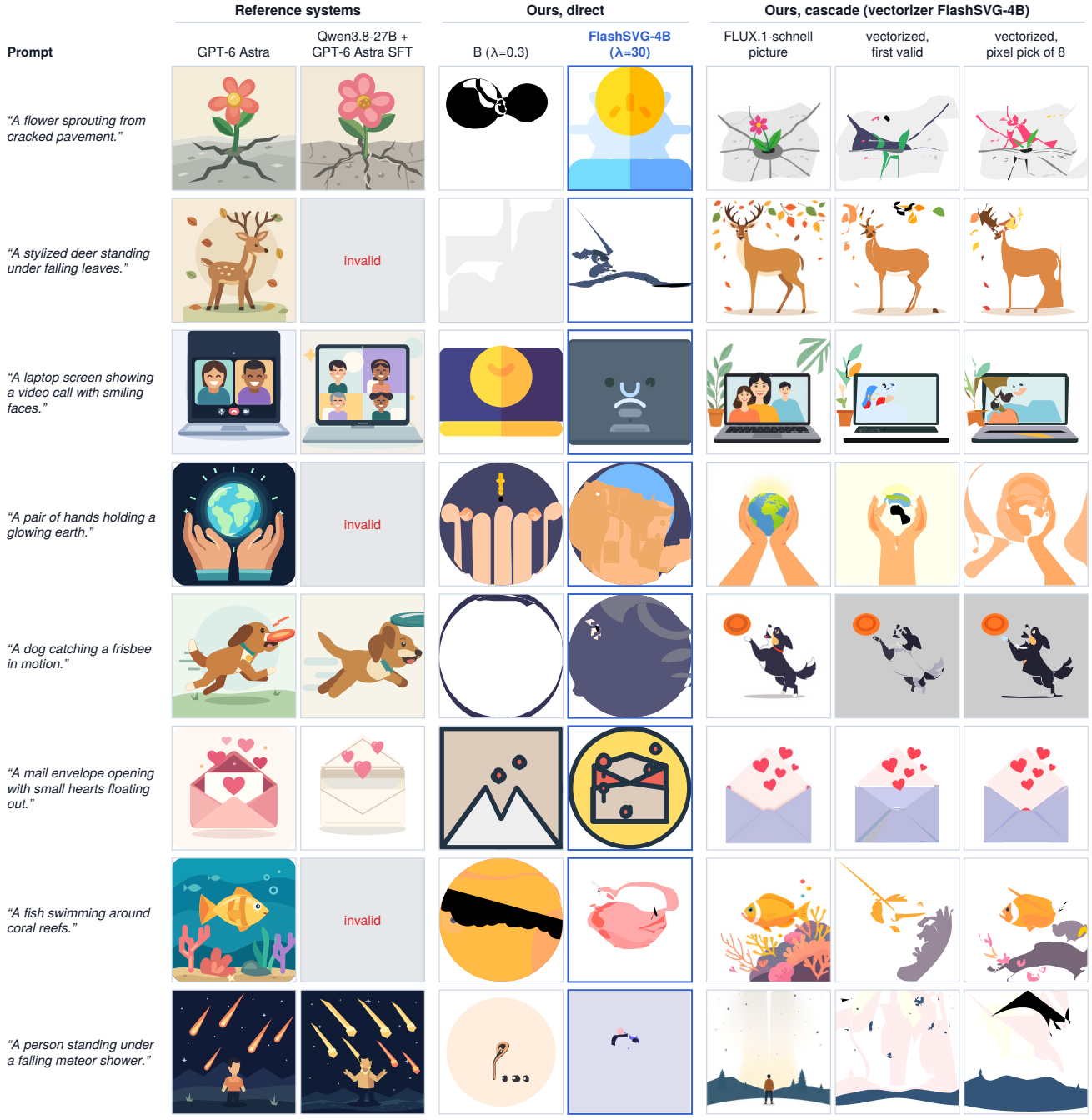


Figure S15. MMSVG-Bench text-to-SVG illustrations, the first eight prompts by id (unselected), with the columns of Figure S14. Grey tiles: no valid SVG.

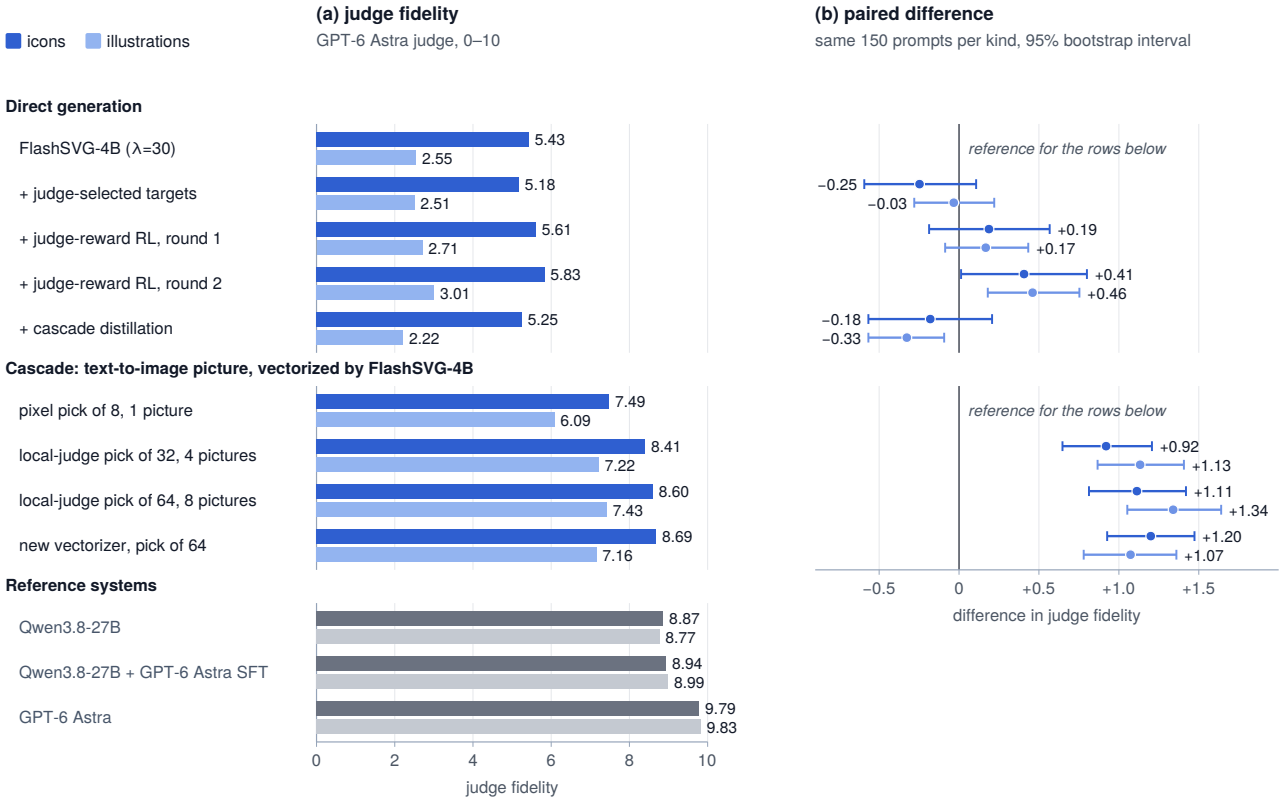


Figure S16. Text-to-SVG judge fidelity on MMSVG-Bench (GPT-6 Astra judge, 0–10; 150 prompts per kind) of direct-generation variants of FlashSVG-4B, of the cascade that draws a text-to-image picture and vectorizes it with FlashSVG-4B, and of three reference systems from Table 1. (a) Means. (b) Paired differences on the same prompts with 95% bootstrap intervals, against FlashSVG-4B for the direct variants and against the pixel pick for the cascade picks. Only the second round of judge-reward RL improves direct generation on both kinds (+0.41 / +0.46); picking cascade candidates with the local judge adds about one point.

Table S10. Transfer of one round of post-training to other model families (image-to-SVG; DINO, first valid of eight, and EOS; icons / illustrations). LoRA unless marked; λ: preference weight. Qwen3.8-27B was post-trained on the 200-unit canvas and is also evaluated on the pixel canvas of Table 1; seed ranges and learning rates in Supplementary Section S6. SArena-600 includes 45 image inputs with a near-duplicate among the training renders (Table S8).

	MMSVG-Bench		SArena-600	
System	**DINO ↑**	**Cand. EOS ↑**	**DINO ↑**	**Cand. EOS ↑**
StarVector-8B [75]				
Released	0.729 / 0.520	0.47 / 0.23	0.797 / 0.125	0.74 / 0.06
λ=0 (targets only)	0.710 / 0.418	0.33 / 0.14	0.805 / 0.136	0.52 / 0.05
λ=1	**0.864** / 0.780	0.88 / 0.79	**0.878** / 0.460	0.94 / 0.38
λ=3	0.841 / 0.798	0.92 / 0.86	0.863 / 0.467	0.95 / 0.43
λ=10 (3 seeds)	0.836 / **0.802**	0.94 / 0.88	0.855 / 0.484	0.96 / 0.52
λ=30	0.814 / 0.785	**0.96 / 0.90**	0.859 / **0.550**	**0.98 / 0.53**
λ=10, full-parameter (best of four LRs)	0.839 / 0.697	0.73 / 0.54	0.863 / 0.344	0.90 / 0.26
Qwen3.8-27B [71], 200-unit canvas				
Base	0.776 / 0.657	0.52 / 0.38	0.868 / 0.173	0.75 / 0.08
λ=10, text-to-SVG replay	**0.911 / 0.905**	0.96 / 0.91	**0.892** / 0.503	0.96 / 0.40
same, full-parameter	0.902 / 0.841	**0.98 / 0.92**	0.838 / **0.645**	**0.98 / 0.92**
Qwen3.8-27B, pixel canvas				
Base	0.887 / 0.809	0.87 / 0.69	–	–
λ=10, text-to-SVG replay	**0.934 / 0.899**	**0.96 / 0.89**	–	–
VinciCoder-7B [114]				
Released	**0.781 / 0.746**	0.92 / 0.78	**0.802** / 0.444	0.90 / 0.35
λ=10, full-param., ground-truth targets (97%)	0.679 / 0.588	0.83 / 0.64	0.721 / 0.348	0.91 / 0.32
λ=10, full-param., own targets (variant)	**0.781** / 0.732	**1.00 / 1.00**	0.794 / **0.597**	**1.00 / 0.99**

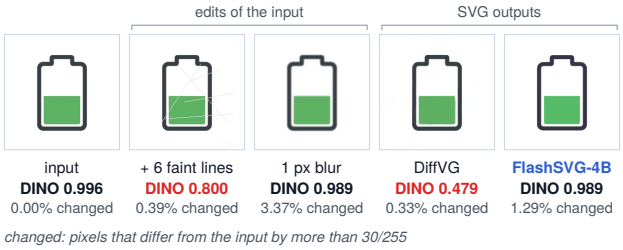


Figure S17. Why DINO under-ranks DiffVG, on one MMSVG-Bench icon: DINO to the input and the share of pixels that differ from it by more than 30/255, for the input, two edits of it, and the rendered outputs of DiffVG (256 paths) and FlashSVG-4B (λ=30). Six faint lines that change 0.39% of the pixels lower DINO to 0.800, while a 1 px blur that changes 3.37% keeps it at 0.989. The DiffVG render changes only 0.33% of the pixels, fewer than FlashSVG-4B (1.29%), yet scores 0.479 against 0.989.

Table S11. Decoding controls on MMSVG-Bench (image-to-SVG candidates that end and DINO, icons / illustrations), each paired against the same model without the rule. "Released" and "Base" rows are the reference runs of the controls (the released OmniSVG-4B gives 0.776 in Table 1); Qwen3.8-27B draws on the 200-unit canvas (Table 1: the input's pixel canvas). ‡Conversion and post-training.

System	Cand. EOS ↑	DINO ↑
OmniSVG-4B [102]		
Released	0.63 / 0.40	0.776 / 0.707
+ ban on repeated 8-grams	0.80 / 0.70	0.730 / 0.620
+ ban on repeated 16-grams	0.72 / 0.52	0.773 / 0.686
+ ban on repeated 32-grams	0.70 / 0.49	0.778 / 0.710
FlashSVG-4B‡	**0.997 / 0.982**	**0.935 / 0.904**
OmniSVG1.1-8B [102]		
Released	0.71 / 0.45	0.800 / 0.690
+ ban on repeated 16-grams	0.81 / 0.62	0.776 / 0.662
Qwen3.8-27B [71]		
Base	0.52 / 0.38	0.776 / 0.657
+ presence penalty 1.5	0.55 / 0.43	0.816 / 0.689
+ post-training (LoRA, replay)	0.96 / 0.91	0.911 / 0.905



Figure S18. Image-to-SVG replies of three API models under the original 200-unit canvas and under the pixel canvas (viewBox 0 0 448 448) on the same five MMSVG-Bench cases, the first in ID order on which all three 200-unit replies leave more than a tenth of the drawing outside the canvas (18–93%). With the 200-unit canvas the models draw in the input's pixel coordinates and the drawing is cropped; none of the 15 pixel-canvas replies is.

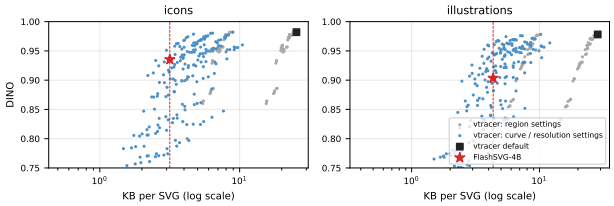


Figure S19. MMSVG-Bench image-to-SVG DINO against mean SVG size for 312 vtracer settings (grey: region settings; blue: curve and tracing-resolution settings; square: defaults, the row of Table 1) and FlashSVG-4B (star; dashed: its mean size). Settings below DINO 0.75 are not shown.

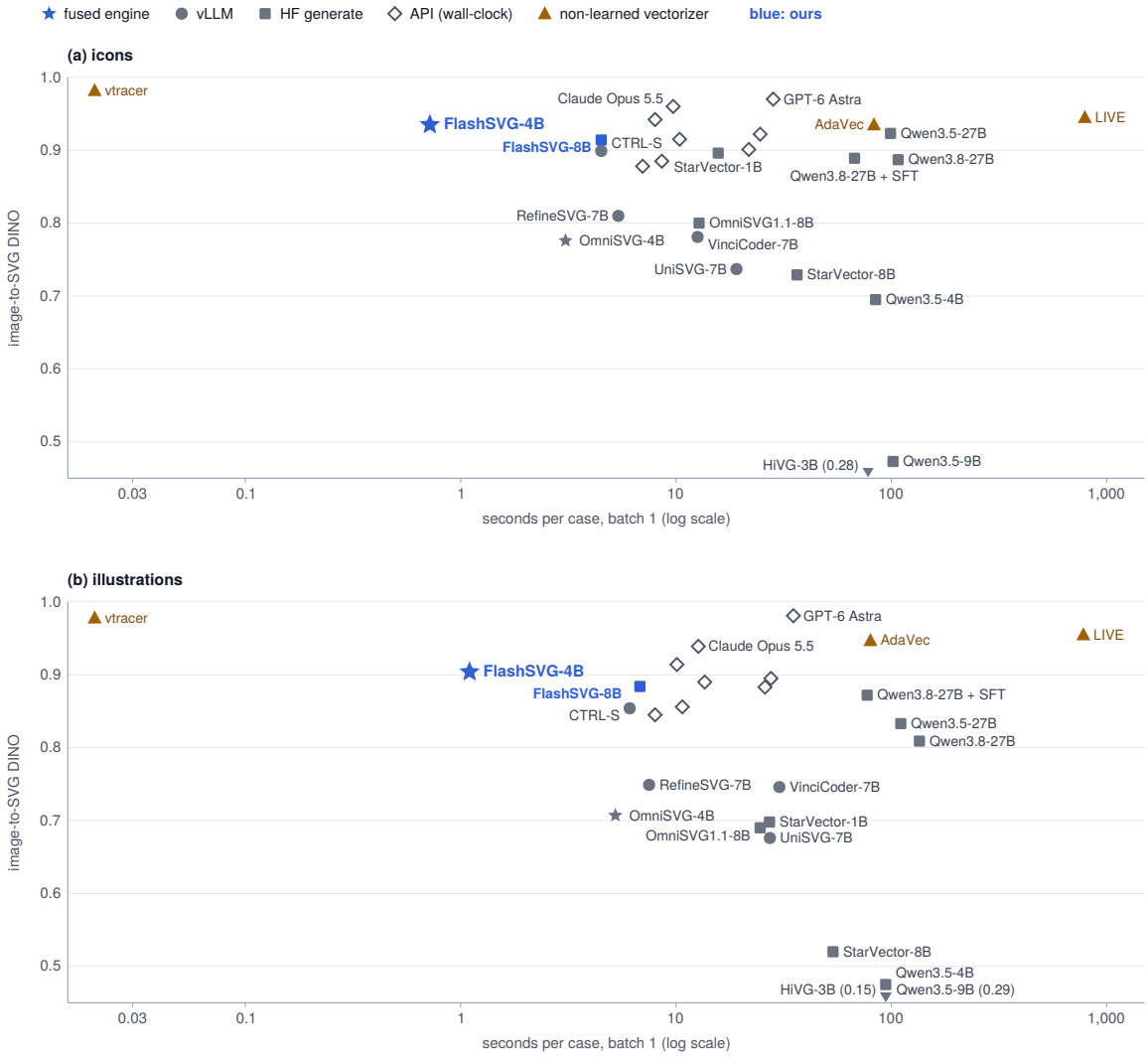


Figure S20. Image-to-SVG DINO (Tables 1 and S8) against batch-1 seconds per case for every timed system, each on the fastest engine we ran (Table S9): (a) icons, (b) illustrations. API models are timed by remote wall-clock, and only GPT-6 Astra and Claude Opus 5.5 among them are labeled (the others are in Table 1); vtracer runs on a CPU. Points below the axis are marked at its bottom edge with their DINO. Not shown: InternSVG-8B (not timed), DiffVG (DINO under-ranks it) and our intermediate rows of Table 1.

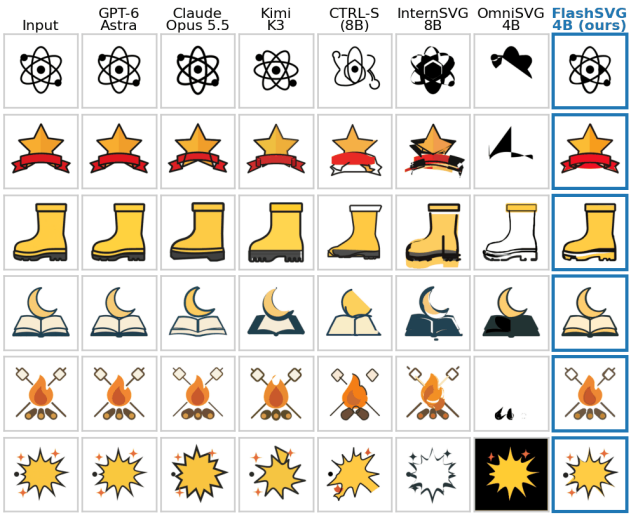


Figure S21. MMSVG-Bench image-to-SVG cases picked by us among the ten per kind where FlashSVG-4B (DINO $\geq$ 0.96) leads the best open learned baseline most (unselected: Figure 4); scored outputs, first valid of eight (API models: one sample).

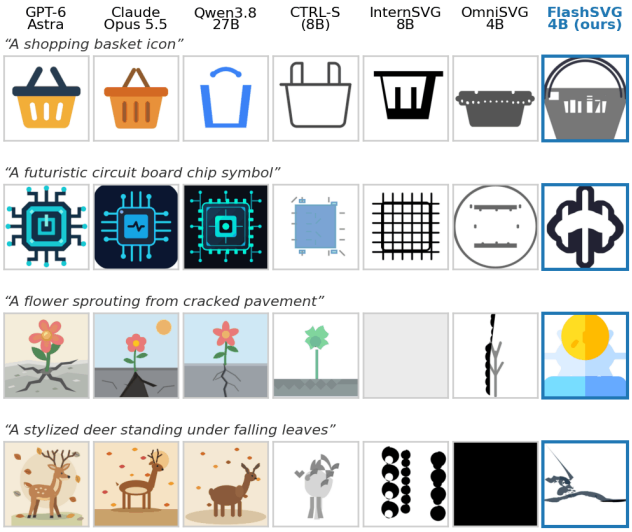


Figure S22. MMSVG-Bench text-to-SVG, the first two icons and illustrations by id (not selected): the scored output of each system (first valid of eight; one sample for API models).



Figure S23. MMSVG-Bench image-to-SVG illustrations, the first eight cases by id (unselected): the input and, per model, its scored output under the official protocol (first valid of eight candidates; HiVG-3B from the authors' inference code, as in Table 1). A grey tile marks a case where none of the eight candidates is a valid SVG.

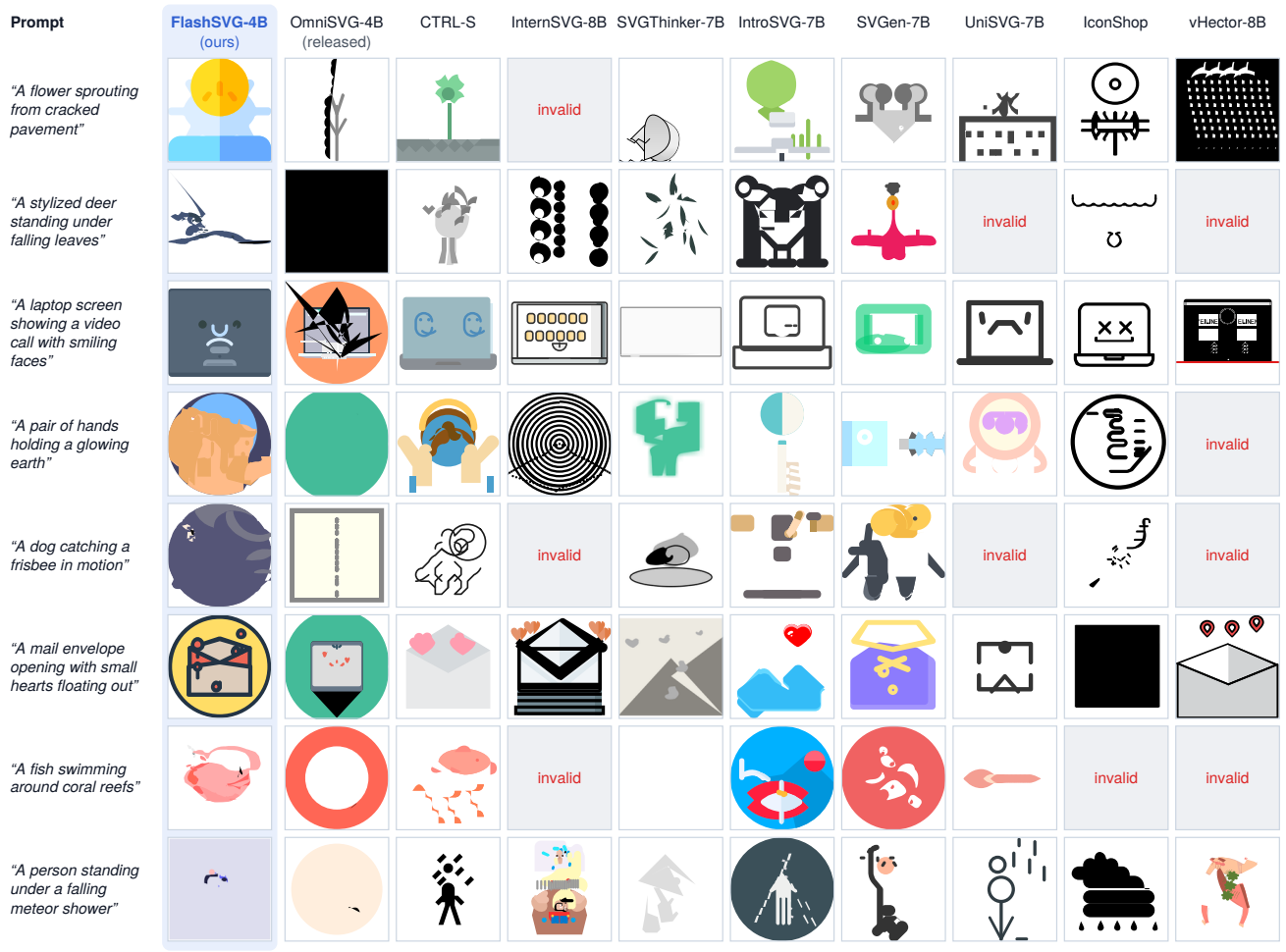


Figure S24. MMSVG-Bench text-to-SVG illustrations, the first eight prompts by id (unselected), prompt on the left; per model, its scored output under the official protocol (first valid of eight candidates). A grey tile marks a prompt where none of the eight candidates is a valid SVG.

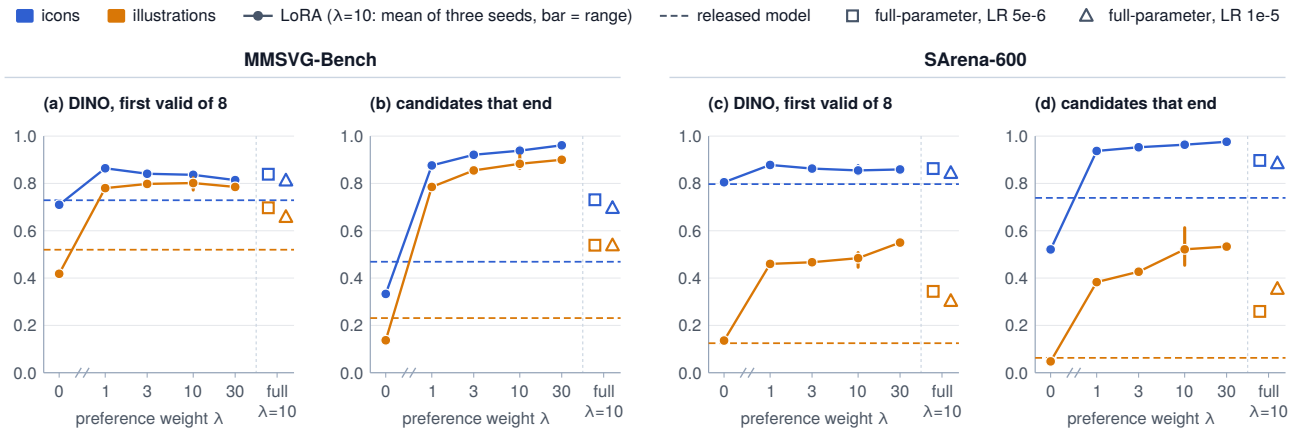


Figure S25. StarVector-8B after one round of termination-aware post-training, on MMSVG-Bench and SArena-600: DINO (first valid of eight) and candidates that end against the preference weight (LoRA; weight 10 with three training seeds, bar = range), with the released model (dashed) and two full-parameter runs at weight 10 (open markers).



Figure S26. StarVector-8B before and after one round of termination-aware post-training (LoRA, λ=10, training seed 1 of 3) on MMSVG-Bench image-to-SVG: the first six icons and the first six illustrations by id (unselected). For each model, the rows show the first of the eight sampled candidates as it is, the first valid of the eight (the scored output of the official protocol), and which of the eight candidates end, in sampling order (filled: ends; outlined: never ends), with their count. A grey tile has nothing to render: "never ends" marks a candidate cut off at the 3,422-token budget, and "no valid SVG" means that none of the eight is valid.

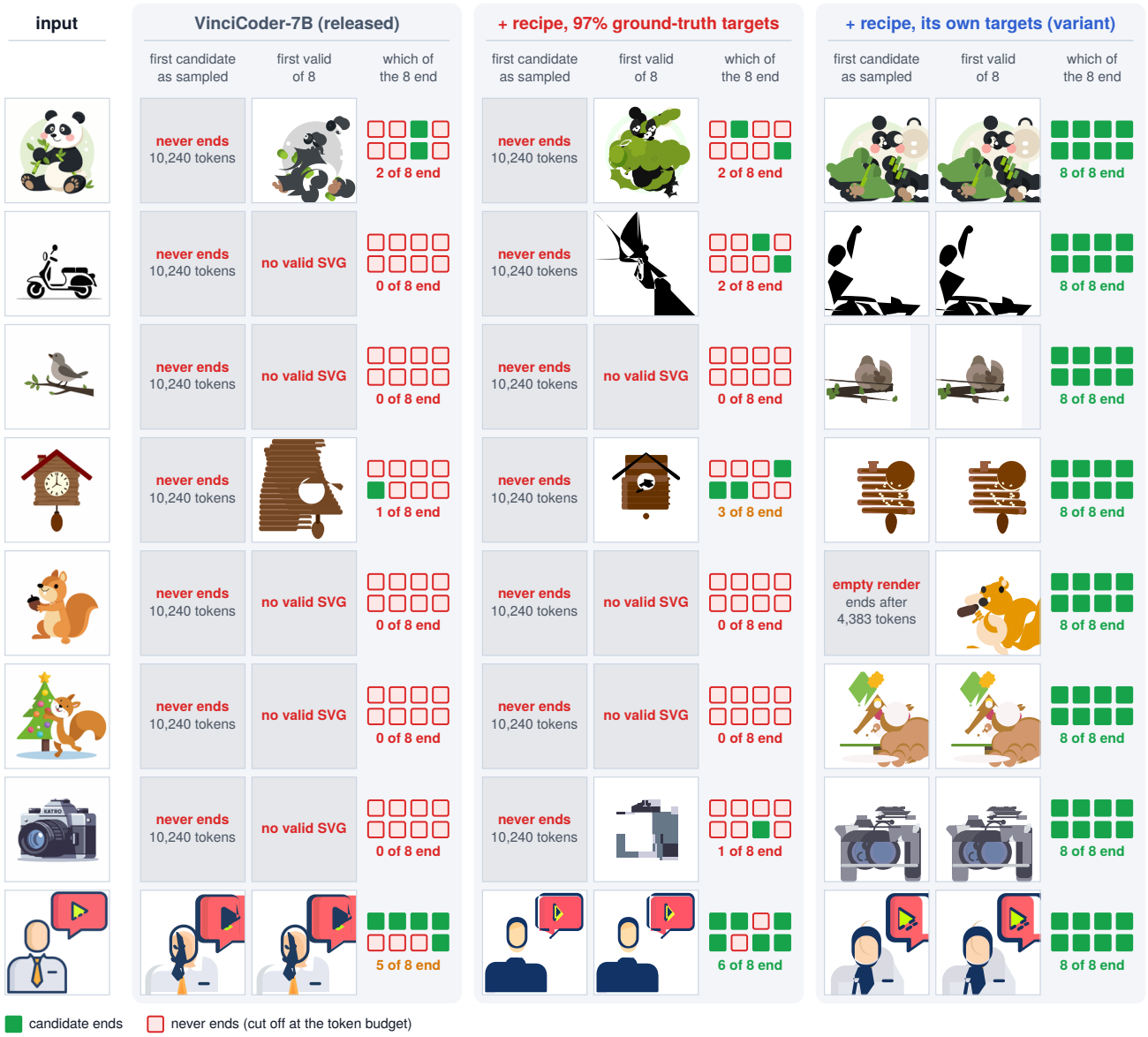


Figure S27. VinciCoder-7B on SArena-600 image-to-SVG illustrations, the first eight by id (unselected): released, and after one round of termination-aware post-training (full-parameter, λ=10, one training seed) with two target rules. Under the recipe's rule, a source's own best candidate is its target only if it reaches the fidelity threshold; VinciCoder's rarely do, so 97% of the targets were ground truth in another style. The variant instead takes each source's best candidate that ends. Per model: the first of the eight sampled candidates as it is, the first valid of the eight (the scored output), and which of the eight candidates end, in sampling order (filled: ends; outlined: never ends), with their count. A grey tile has nothing to render: "never ends" marks a candidate cut off at the 10,240-token budget, "empty render" one that ends but draws nothing, and "no valid SVG" means that none of the eight is valid.

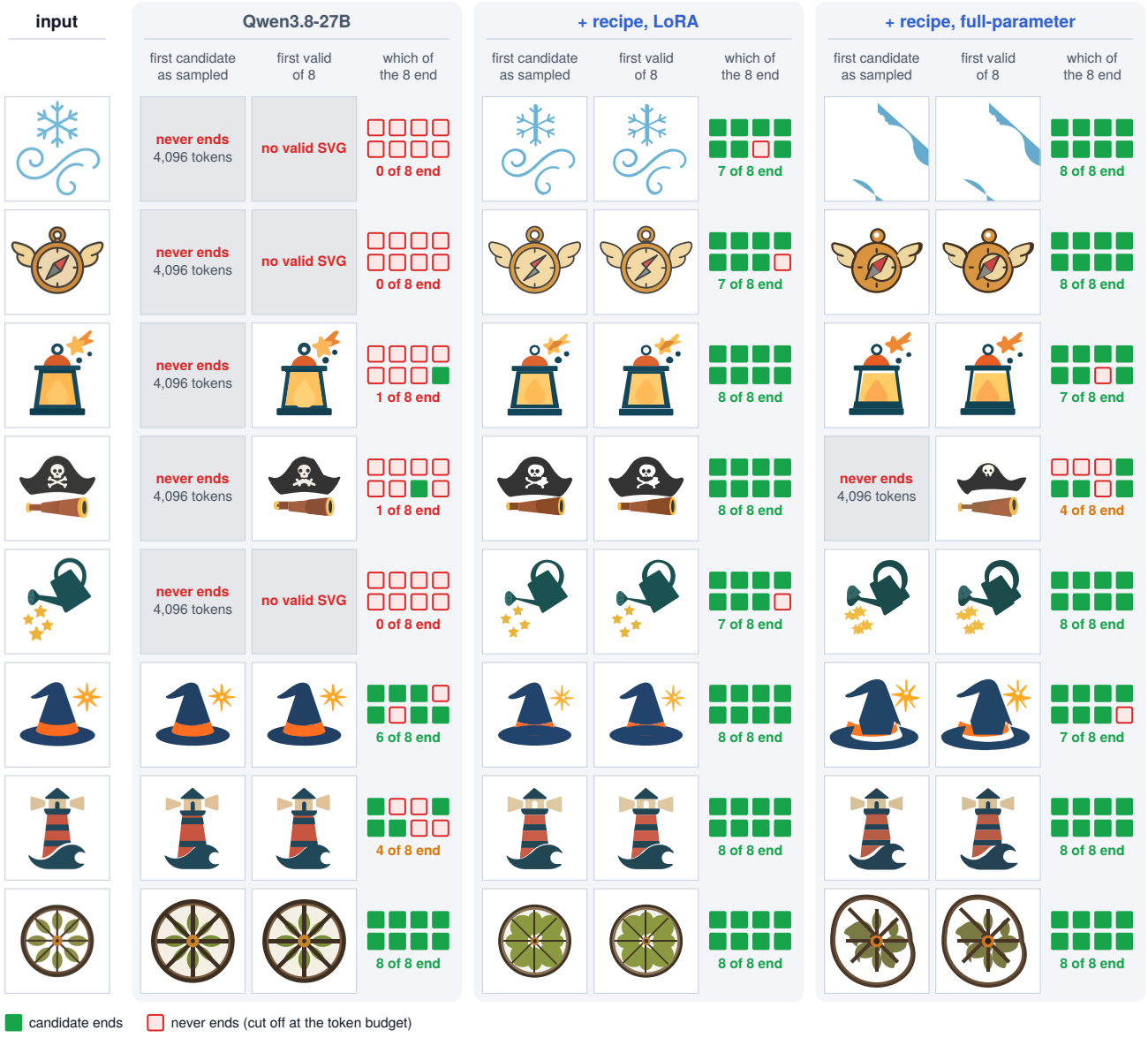


Figure S28. Qwen3.8-27B on MMSVG-Bench image-to-SVG illustrations, the first eight by id (unselected), on the 200-unit canvas: base, and after one round of termination-aware post-training (λ=10 with text-to-SVG replay, training seed 1) with LoRA and with full-parameter updates on the same data. Per model: the first of the eight sampled candidates as it is, the first valid of the eight (the scored output), and which of the eight candidates end, in sampling order (filled: ends; outlined: never ends), with their count. A grey tile has nothing to render: "never ends" marks a candidate cut off at the 4,096-token budget, and "no valid SVG" means that none of the eight is valid.

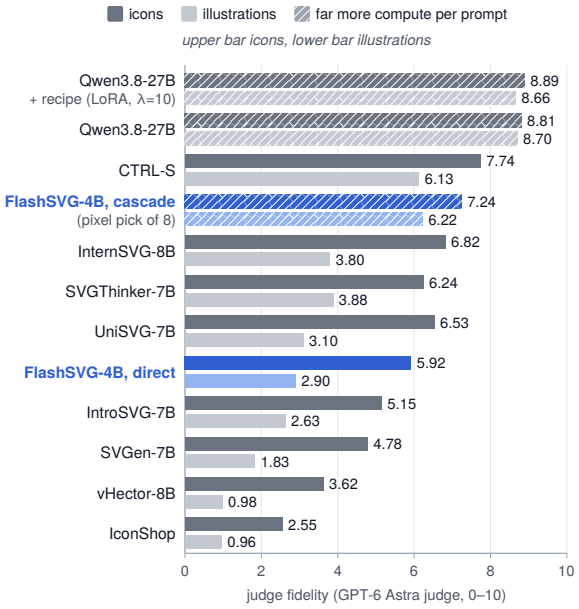


Figure S29. SArena-600 text-to-SVG judge fidelity (GPT-6 Astra judge, 0–10; first valid of eight; upper bar icons, lower bar illustrations). The hatched rows, the cascade and the two Qwen3.8-27B models, cost far more compute per prompt.